

Tesis de Doctorado en Ingeniería

Generative modeling for the synthesis of vascular anatomical structures

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Prefacio

Esta Tesis es presentada como parte de los requisitos para optar al grado académico de Doctora en Ingeniería, de la Universidad Nacional del Sur, y no ha sido presentada previamente para la obtención de otro título en esta Universidad u otras. La misma contiene los resultados obtenidos en investigaciones llevadas a cabo en el Laboratorio de Inteligencia Artificial de la Universidad Torcuato Di Tella durante el período comprendido entre el día 1 de Abril de 2021 y el 1 de Septiembre de 2025, bajo la dirección del Dr. Emmanuel Iarussi, profesor investigador de la escuela de negocios de la Universidad Torcuato Di Tella e Investigador Adjunto del CONICET, y bajo la codirección del Dr. Claudio Delrieux, profesor titular del Departamento de Ingeniería Eléctrica y de Computadoras e Investigador Principal del CONICET.

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La presente tesis ha sido aprobada el	
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Resumen

Las enfermedades cardiovasculares son la principal causa de muerte a nivel mundial, y el modelado preciso de los vasos sanguíneos es fundamental para desarrollar herramientas clínicas de diagnóstico, planificación del tratamiento y simulación. A pesar de la creciente disponibilidad de datos de imágenes médicas, la generación de geometrías vasculares diversas y realistas sigue siendo un desafío debido a la complejidad estructural y la variabilidad anatómica de los árboles vasculares. Esta tesis aborda estos desafíos introduciendo dos modelos generativos profundos para la síntesis de estructuras tridimensionales de vasos sanguíneos.

La primera contribución, VesselVAE, es un autoencoder variacional recursivo que codifica y decodifica árboles vasculares representados como estructuras binarias, propagando información desde las hojas hasta la raíz y generando subárboles de manera jerárquica en la decodificación. El modelo aprende de forma conjunta topología (presencia/ausencia de ramas y patrones de ramificación) y geometría (posiciones, radios, longitudes), mediante un espacio latente que captura dependencias de largo alcance a nivel de subárbol. Una vez entrenado es capaz de generalizar entre distintos conjuntos de datos, incluidos aquellos con casos patológicos como aneurismas, superando a los métodos existentes en métricas cuantitativas estándar.

La segunda contribución, VesselGPT, es un modelo autorregresivo que sintetiza estructuras vasculares como secuencias de tokens aprendidos. Introduce una representación discreta y compacta de los vasos mediante un codebook aprendido y modela las formas de las secciones transversales usando B-splines. Su formulación secuencial permite la generación condicional, habilitando una mayor expresividad anatómica y manteniendo la eficiencia computacional. En conjunto, estos modelos avanzan el estado del arte en la síntesis

de geometría vascular 3D, combinando precisión, flexibilidad anatómica y eficiencia. Proporcionan una base para el modelado vascular impulsado por datos con aplicaciones potenciales en la medicina computacional.

En conjunto, estos modelos avanzan el estado del arte en la síntesis de geometría vascular 3D al combinar precisión, flexibilidad anatómica y eficiencia. Ofrecen una base sólida para el modelado vascular basado en datos con amplias aplicaciones en medicina computacional.

Abstract

Cardiovascular diseases are the leading cause of death worldwide, and accurate modeling of blood vessels is essential for developing clinical tools for diagnosis, treatment planning, and simulation. Despite the growing availability of medical imaging data, generating diverse and realistic vascular geometries remains challenging due to the structural complexity and anatomical variability of vascular trees. This thesis addresses these challenges by introducing two deep generative models for the synthesis of three-dimensional blood vessel structures.

The first contribution, VesselVAE, is a recursive variational autoencoder that encodes and decodes vascular trees represented as binary structures, propagating information from the leaves to the root and generating subtrees hierarchically during decoding. The model jointly learns topology (presence/absence of branches and branching patterns) and geometry (positions, radii, lengths) through a latent space that captures long-range dependencies at the subtree level. Once trained, it is capable of generalizing across different datasets, including those with pathological cases such as aneurysms, outperforming existing methods on standard quantitative metrics.

The second contribution, VesselGPT, is an autoregressive model that synthesizes vascular structures as sequences of learned tokens. It introduces a discrete, compact representation of vessels via a learned codebook and models cross-sectional shapes using B-splines. Its sequential formulation enables conditional generation, allowing greater anatomical expressiveness while maintaining computational efficiency. Together, these models advance the state of the art in 3D vascular geometry synthesis by combining accuracy, anatomical flexibility, and efficiency. They provide a foundation for data-driven vascular modeling with potential applications in computational medicine.

Together, these models advance the state of the art in 3D vascular geometry synthesis by combining accuracy, anatomical flexibility, and efficiency. They offer a solid foundation for data-driven vascular modeling with broad applications in computational medicine.

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Abbreviations & Acronyms

1-NNA 1-Nearest Neighbor Accuracy.

3D three-dimensional.

AAA abdominal aortic aneurysms.

AVMs Arteriovenous malformations.

CAD Coronary Artery Disease.

CCO Constrained Constructive Optimization.

CD Chamfer Distance.

CFD Computational Fluid Dynamics.

CNNs convolutional neural networks.

Cov Covergae.

CoW Circle of Willis.

CS Cosine Similarity.

CT Computed Tomography.

CTA Computed Tomography Angiography.

DNNs Deep Neural Networks.

EMD Earth Mover's Distance.

FEM finite element methods.

GANs Generative Adversarial Networks.

MMD Minimum Matching Distance.

MRA Magnetic Resonance Angiography.

MRI Magnetic Resonance Imaging.

OSI Oscillatory Shear Index.

PCI Percutaneous Coronary Intervention.

ReLU Rectified Linear Unit.

RNN Recursive Neural Network.

RvNN Recursive Neural Network.

SDFs Signed Distance Functions.

tanh Hyperbolic Tangent.

VAE Variational Autoencoder.

VQ-VAE Vector-Quantized Variational Autoencoder.

WSS Wall Shear Stress.

Chapter 1

Introduction

Vascular diseases remain a leading focus of medical research due to their widespread prevalence and significant mortality. Cardiovascular diseases alone accounted for approximately 17.9 million deaths in 2019—roughly 32% of global fatalities—with over 75% occurring in low- and middle-income countries [12]. In Argentina, an estimated 7.5% of the population was affected by CVDs in 2021. The most prevalent conditions include:

- **Coronary Artery Disease (CAD):** The leading cause of ischemic heart disease, CAD was responsible for 7 million deaths in 2010, rising to 9 million by 2021, leading cause of death worldwide [76], [104]. Globally, over 315 million people are currently living with CAD. In Argentina alone, 57,257 deaths were attributed to ischemic heart disease in 2021.
- **Stroke (cerebrovascular disease):** Encompassing a range of disorders affecting the brain’s vasculature—including aneurysms and strokes—cerebrovascular disease ranks as the second most fatal cardiovascular disease, with 21,675 related deaths reported in Argentina in 2021 [83].

As computational modeling becomes increasingly central to medical research and clinical workflows, the ability to generate accurate and diverse three-dimensional (3D) vascular representations will continue to be critical for improving diagnosis, treatment, and understanding of cardiovascular diseases. Given their life-threatening consequences and the global burden of these conditions, accurate modeling of the vascular system is essential for advancing both research and clinical practice, with the potential for significant impact

on global healthcare outcomes. Virtual 3D anatomical models play a key role in this endeavor. These models facilitate in-depth analysis of vascular geometry and hemodynamics, enabling a wide range of applications such as in-silico experimentation, image-guided diagnostics, predictive modeling, treatment planning, and medical education, among others [45], [46], [58], [81], [102]. In vascular medicine, 3D models of arteries, veins, and microvascular networks are essential for understanding blood flow dynamics, simulating physiological functions, and planning interventions. The typical workflow begins with high-resolution volumetric imaging, such as Computed Tomography Angiography (CTA) or Magnetic Resonance Angiography (MRA), followed by vessel segmentation and subsequent reconstruction into surface meshes or centerline graphs [28], [37], [52], [93]. These anatomical representations are frequently employed in Computational Fluid Dynamics (CFD) simulations to quantify clinically relevant parameters, including pressure gradients, flow velocities, wall shear stress, and regions of recirculation or stagnation [22], [59], [65], [97], [125].

Virtual vascular models are used in a wide range of applications, including in-silico experimentation, image-guided diagnostics, patient-specific treatment planning, and medical education [46], [58], [81]. They also underpin non-invasive simulations of hemodynamics through computational fluid dynamics (CFD), enabling the quantification of clinically relevant parameters such as blood flow velocity, pressure gradients, and wall shear stress [22], [59], [65]. These simulations provide insight into disease progression and support decisions about interventions, including stent placement or aneurysm repair [18], [106].

Despite recent advancements in medical imaging and segmentation algorithms, generating high-quality vascular models remains challenging. The standard pipeline typically involves high-resolution imaging (e.g., CTA or MRA), followed by vessel enhancement, segmentation, and 3D reconstruction into surface meshes or centerline graphs [52], [93]. However, each of these steps can introduce inaccuracies, especially when imaging small-caliber or tortuous vessels [2], [50]. Moreover, many simulation pipelines rely on empirical assumptions—such as Murray’s law [57] or default boundary conditions—that may not reflect patient-specific physiology, limiting their realism and clinical utility [80], [109].

In addition to technical limitations, the availability of high-quality, annotated vascular data is severely constrained. Acquiring and labeling patient scans is time-consuming, costly, and often restricted by privacy regulations. This issue is even more pronounced for rare diseases, pediatric populations, or conditions with low prevalence, where curated

datasets are especially scarce [111]. These barriers hinder the development, evaluation, and reproducibility of data-driven methods in vascular modeling.

In response to these challenges, the use of synthetic data has emerged as a promising alternative. Generative models such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion models allow researchers to simulate anatomically realistic vascular structures from limited training data while avoiding privacy concerns. These techniques have demonstrated success in other domains of medical imaging—e.g., generating synthetic liver lesions for classification [27], brain MRIs and CT scans with diffusion models [39], and synthetic lung nodules in CT via conditional GANs [32]. Their application to vascular modeling has the potential to accelerate algorithm development and improve performance, particularly in data-scarce settings [9], [11].

However, existing generative methods for vascular structure synthesis often face a trade-off between topological accuracy, geometric realism, and compatibility with downstream tasks such as simulation or segmentation. For example, graph-based models may capture topology but lack surface fidelity, while voxel- or SDF-based methods excel at geometry but do not preserve connectivity [42], [69], [90]. Bridging this gap remains an open problem in the field.

In this thesis, two novel methods are proposed for the synthesis of three-dimensional vascular geometries, addressing the problem from deep learning-based generative approaches. The first, *VesselVAE*, is a generative framework that employs a Recursive Neural Network (RvNN) to model the hierarchical structure of vascular trees. This architecture, successfully applied in domains such as natural language processing and semantic shape modeling, enables the capture of both topological connectivity and local geometric features of vessels through the construction of a low-dimensional latent space. Once trained, the model can be sampled to generate new vascular geometries that preserve essential properties such as radius, length, and tortuosity—even in cases with malformations such as aneurysms. To the best of our knowledge, this work represents the first application of RvNNs for blood vessel synthesis. Moreover, we conducted an extensive evaluation on both real and synthetic datasets, demonstrating the model’s ability to generate plausible and anatomically coherent structures.

As a second contribution, this thesis presents *VesselGPT*, a novel approach based on autoregressive Transformers, which addresses vascular synthesis as a sequential modeling

problem. While Transformers have shown remarkable performance in tasks involving sequential and geometric structures, their application to vascular tree generation remains scarce. In this work, we propose a representation of vessels as a sequence of nodes discretized through a vector-quantized autoencoder (VQ-VAE), which captures geometric patterns in a compact vocabulary. This vocabulary is then used to train a GPT-2 model capable of autoregressively generating complex vascular structures. Additionally, we introduce a cross-sectional representation based on B-splines to more accurately model the vessel lumen shape, overcoming the limitation of assuming a single radius value. We validate the effectiveness of this approach through exhaustive comparisons with previous methods.

1.1 Objectives of the thesis

1.1.1 General objective

To develop and evaluate generative models for synthesizing realistic 3D vascular structures using deep learning techniques, enabling plausible generation of anatomical geometries.

1.1.2 Specific objectives

- To design and implement a recursive variational autoencoder that encodes vascular structures as binary trees and reconstructs them by learning hierarchical geometric and topological features.
- To develop an autoregressive transformer-based model for sequential generation of vascular structures, through tokenized representations.

1.1.3 Structure of the thesis

This thesis comprises six chapters, each addressing a specific stage of the research. Chapter 1 introduces the research problem, its scientific and practical relevance, and the specific objectives of the work. Chapter 2 presents the conceptual framework, providing the theoretical foundations and a detailed description of the methods and algorithms employed. Chapter 3 focuses on the design and development of a generative modeling approach for vascular structures based on recursive neural networks, highlighting its principles and implementation. Chapter 4 builds on the previous chapter, benchmarking the proposed

method against baseline approaches and conducting ablation studies across datasets and data characteristics. Chapter 5 introduces a novel autoregressive methodology for vascular structure modeling, detailing its motivation, architecture, and contributions relative to existing methods.

Finally, Chapter 6 summarizes the main conclusions, discusses their implications, and outlines avenues for future work. The content of Chapter 3 is based on Publications 1 and 2; Chapter 4 corresponds to Contribution 3; and Chapter 5 builds upon Publication 4 and an additional manuscript currently in preparation.

1.2 Contributions

1.2.1 Publications in Peer-Reviewed Journals and Conferences

1. **Feldman, P.**, Fainstein, M., Siless, V., Delrieux, C., & Iarussi, E. (2023, October). VesselVAE: Recursive variational autoencoders for 3D blood vessel synthesis. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 67-76). Cham: Springer Nature Switzerland.
2. **Feldman, P.**, Fainstein, M., Siless, V., Delrieux, C., & Iarussi, E. (2024). Modelado generativo de estructuras vasculares 3D *LIII Jornadas Argentinas de Informática (JAIIO), Bahía Blanca, Argentina, 12-16 de agosto de 2024. I.*
3. **Feldman, P.**, Fainstein, M., Siless, V., Delrieux, C., & Iarussi, E. (2025, June). Recursive variational autoencoders for 3D blood vessel synthesis. *Medical Image Analysis*.
4. **Feldman, P.**, Sinnona, M., Delrieux, C., Siless, V. & Iarussi, E. (2025, September). VesselGPT: Autoregressive Modeling of Vascular Geometry. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 662–672). Cham: Springer Nature Switzerland.

Highlighted for oral presentation

Chapter 2

Background

2.1 Human vascular network

The human vascular system is a highly organized and extensive network responsible for the distribution of oxygen and nutrients throughout the body. It is composed of arteries, veins, and capillaries, each fulfilling specific physiological roles[60], [94]. Blood circulates through two interconnected systems: the systemic and pulmonary circulations. These systems are composed of a vast network of blood vessels, broadly classified as arteries and veins. Arteries transport blood away from the heart, whereas veins return blood toward the heart. In the systemic circulation, oxygen-rich blood is pumped from the heart into the aorta, which branches into major arteries such as the hepatic artery (supplying the liver) and the renal arteries (supplying the kidneys). As these arteries approach their target organs, they progressively divide into smaller vessels. The smallest arterial branches, known as arterioles, further subdivide into capillaries, where the exchange of gases, nutrients, and metabolic waste occurs between blood and surrounding tissues.

Following this exchange, deoxygenated blood begins its return journey via small veins called venules, which gradually merge into larger veins. Within each organ, this venous network culminates in major veins—such as the hepatic and renal veins—that drain into the inferior vena cava, directing blood back to the heart. In contrast, the pulmonary circulation begins with the ejection of deoxygenated blood from the heart through the pulmonary arteries toward the lungs. There, gas exchange occurs in the pulmonary capillaries, and the now oxygenated blood returns to the heart via the pulmonary veins,

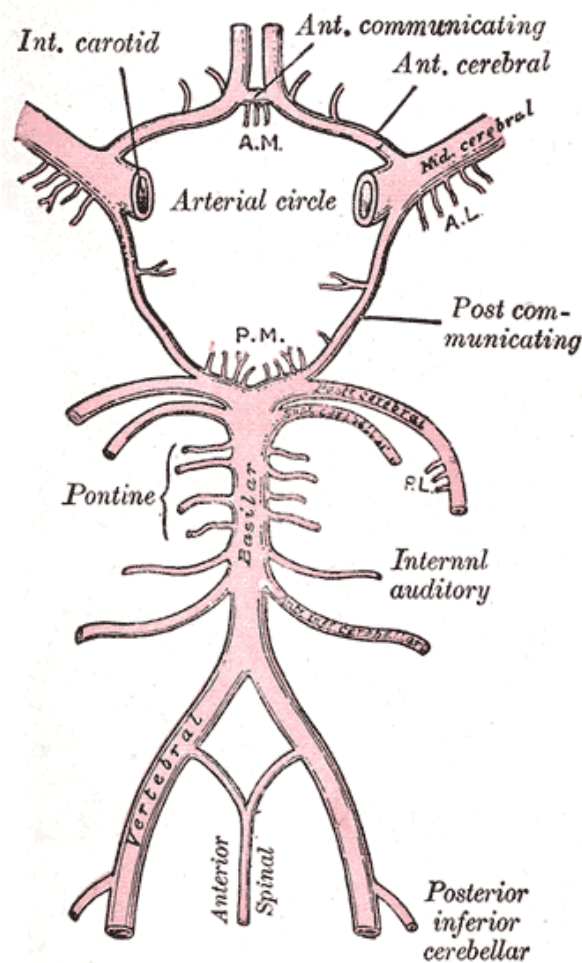


Figure 2.1: Schematic diagram of the Circle of Willis and its major branches[94].

completing the circuit.

An additional key subsystem is the cerebral circulation, responsible for supplying oxygen and nutrients to the brain. This system begins with the internal carotid and vertebral arteries, which converge to form the Circle of Willis (CoW)—a circular arterial structure at the base of the brain that provides redundant pathways for blood flow (Fig. 2.1). From the Circle of Willis arise the anterior, middle, and posterior cerebral arteries, which branch extensively to perfuse different brain regions. Venous drainage of the brain occurs through a network of cerebral veins and dural venous sinuses, ultimately emptying into the internal jugular veins.[70].

Among the most clinically significant vascular pathologies are aneurysms: localized

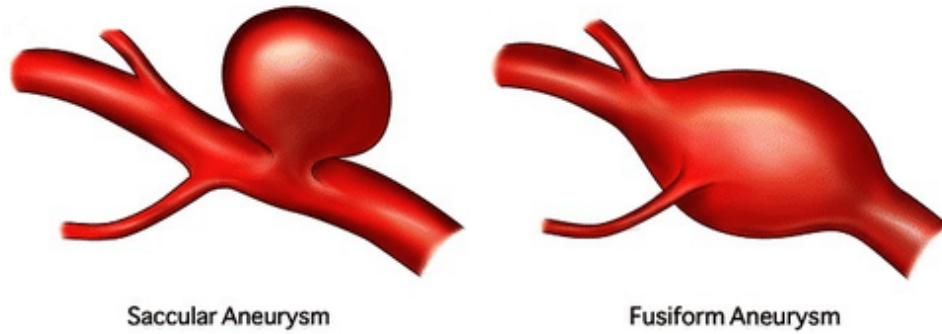


Figure 2.2: Illustration of saccular and fusiform aneurysms. Saccular aneurysms, are characterized by a localized outpouching on one side of the vessel wall and are most commonly found in the cerebral vasculature. Fusiform aneurysms involve circumferential dilation of the entire vessel wall and are more frequently associated with large arteries such as the abdominal aorta [35].

dilations of the arterial wall that typically arise due to structural weaknesses, often compounded by genetic predisposition, chronic hypertension, or mechanical stress. Aneurysms can occur in several vascular territories, most notably within the cerebral circulation (intracranial aneurysms) and the abdominal aorta (Abdominal aortic aneurysms (AAA)) (Fig. 2.2). Intracranial aneurysms are frequently located at arterial bifurcations in the Circle of Willis, with the anterior communicating artery being a common site. These aneurysms often adopt a saccular (berry-like) morphology and pose a significant risk of rupture, leading to subarachnoid hemorrhage. In contrast, AAA are typically fusiform in shape and develop below the renal arteries. They are commonly detected through imaging and are defined by a diameter exceeding 3 cm. Rupture of an AAA is a medical emergency, often associated with high mortality rates. From a modeling perspective, aneurysms pose considerable challenges due to their irregular shapes and the need for high-fidelity geometric representation. These structures deviate significantly from idealized, cylindrical vessel models and demand approaches that can accurately capture their variable morphology.

In addition to aneurysms, several other vascular pathologies significantly impact vessel morphology and function, and are therefore relevant for both clinical practice and computational modeling. *Atherosclerosis* is a chronic degenerative disease characterized by the accumulation of lipid-rich plaques within the arterial wall, leading to luminal narrowing,

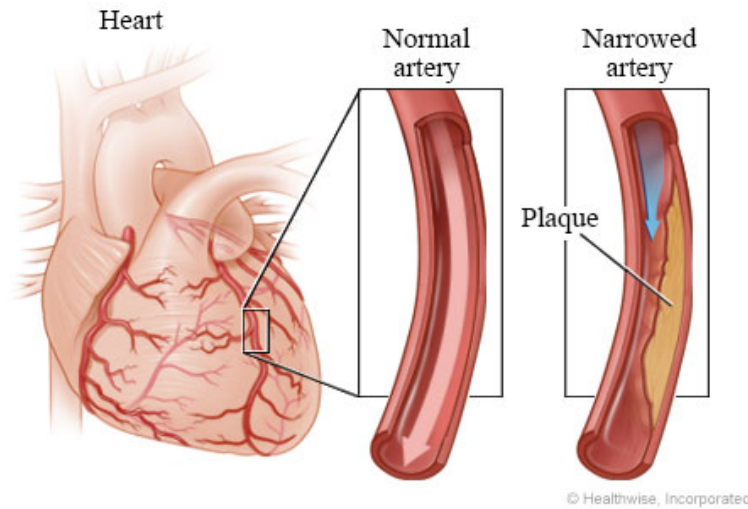


Figure 2.3: Illustration of the coronary arteries and a cross-sectional view of an artery with and without atherosclerotic plaque. The buildup of plaque in the arterial wall leads to narrowing of the lumen, reduced blood flow, and an increased risk of cardiovascular events such as myocardial infarction. This figure highlights both the anatomical location of the coronary arteries and the pathological impact of atherosclerosis on vascular integrity. Source: reproduced from “Thoracic Aortic Aneurysm”, University of Vermont Health Network [105].

wall thickening, and reduced blood flow. It commonly affects coronary, carotid, and peripheral arteries, and is a major contributor to cardiovascular and cerebrovascular events [51]. A related manifestation is *arterial stenosis*, which refers to the localized narrowing of a vessel, often resulting from advanced atherosclerotic plaque development. Stenosis alters hemodynamic patterns, induces turbulence, and increases pressure gradients across the affected segment, factors that are critical in both diagnosis and simulation-based assessment [96] (Fig. 2.3).

Arteriovenous malformations (AVMs) are congenital anomalies in which abnormal, direct connections form between arteries and veins, bypassing the capillary bed. These structures disrupt the typical tree-like topology of the vasculature and can lead to high-flow shunting, increased risk of hemorrhage, and complications such as seizures or neurological deficits when located in the brain [86].

Another important context in which vascular morphology deviates significantly from healthy anatomy is in the presence of cancerous tumors. Solid tumors stimulate the

formation of new blood vessels through a process known as *tumor-induced angiogenesis* [25]. However, unlike normal vasculature, tumor-associated vessels tend to be highly disorganized, tortuous, and leaky, with irregular branching and heterogeneous diameters. These abnormalities impair efficient blood flow and oxygen delivery, contributing to tumor progression, metastasis, and resistance to therapies [15].

2.2 Clinical Motivation and Applications of 3D Vascular Models

Cardiovascular diseases (CVDs) are still the foremost cause of death worldwide, responsible for almost 18 million deaths in 2019, over 75% of which occurred in low- and middle-income countries [12]. In Argentina, CVDs were estimated to affect 7.5% of the population in 2021, with ischemic heart disease and stroke as the main contributors. In light of this global burden, improving our understanding, diagnosis, and treatment of vascular pathologies remains a critical priority for both medicine and research.

Three-dimensional (3D) models of arteries, veins, and microvascular networks play a pivotal role in this effort. These models are essential for studying blood flow dynamics, simulating physiological functions, and supporting both pre-operative planning and post-operative evaluation [46], [58], [81]. Applications include aneurysm treatment planning, stent selection, device design, and longitudinal assessment of disease progression.

A typical 3D vascular modeling pipeline starts with high-resolution volumetric imaging—most often computed tomography angiography (CTA) or magnetic resonance angiography (MRA)—and proceeds through segmentation and geometric reconstruction into either surface meshes or centerline-based graphs [28], [37], [52], [93]. These geometric representations provide the basis for computational fluid dynamics (CFD) simulations, which in turn enable the estimation of clinically relevant hemodynamic quantities, including pressure gradients, velocity fields, Wall Shear Stress (WSS), Oscillatory Shear Index (OSI), and recirculation zones [22], [59], [65], [97], [125].

2.3 From Imaging to Geometry: The Reconstruction Pipeline

Generating accurate three-dimensional (3D) vascular models from clinical imaging data involves a multi-stage pipeline, each step critical in preserving anatomical integrity and enabling reliable downstream analyses [52], [124]:

1. **Vessel enhancement:** Early processing often applies multi-scale vesselness filters, such as the Frangi filter, to accentuate tubular structures while suppressing irrelevant background features [26]. Learnable enhancements via convolutional neural networks (CNNs) have also shown promise in refining vasculature in Computed Tomography (CT) images [122].
2. **Segmentation:** Vascular segmentation has transitioned from classical model-based approaches like active contours [40] to modern deep learning frameworks such as U-Net [82] and transformer-augmented architectures demonstrating strong performance in cerebrovascular segmentation tasks [124].
3. **Geometric reconstruction:** Once a binary or probabilistic segmentation mask is obtained, vascular geometry can be reconstructed in two primary formats:
 - *Centerline graphs*, where a skeletonized representation of the vascular tree is extracted to capture the branching topology and local radii. These are commonly generated using topological thinning [47], or medial axis transforms [99]. Popular open-source libraries like the Vascular Modeling Toolkit (VMTK) [5] provide streamlined implementations for centerline extraction and analysis.
 - *Surface meshes*, typically obtained via isosurface extraction algorithms such as Marching Cubes [54]. These triangulated meshes accurately capture the vessel lumen boundaries and are widely used in computational fluid dynamics (CFD) and finite element modeling workflows. Mesh quality can be improved post hoc using remeshing or Laplacian smoothing techniques [10].
4. **Post-processing and quality assurance:** Refinement commonly involves removing spurious branches, smoothing surfaces, remeshing to improve triangle quality, enforcing topological consistency, and ensuring watertight geometry for computational fluid dynamics (CFD) readiness.

Despite improvements, challenges persist: limited imaging resolution frequently omits small-caliber vessels; motion artifacts and noise lead to segmentation errors; manual intervention remains necessary in complex or pathological cases; and systemic errors can propagate through to simulations—compromising clinical utility.

2.4 Vascular Simulations

Once anatomical vascular models are reconstructed—typically as centerline graphs or watertight surface meshes—they can be used as computational domains for simulating blood flow via computational fluid dynamics (CFD). These simulations solve the incompressible Navier–Stokes equations under physiological boundary conditions to characterize hemodynamic behavior within the vascular network [102], [106].

Blood is commonly modeled as an incompressible Newtonian fluid in medium-to-large arteries, where shear rates are sufficiently high for viscosity to remain approximately constant. However, in smaller vessels—such as cerebral arterioles, retinal capillaries, or coronary side branches—non-Newtonian effects become significant. In such cases, shear-thinning models (e.g., Carreau–Yasuda, Casson) are used to capture the dependence of blood viscosity on local flow velocity [68], [85].

Boundary Conditions and Modeling Assumptions

To approximate physiological conditions, boundary conditions must be carefully chosen:

- **Inlet conditions** often involve time-resolved (pulsatile) flow or velocity profiles measured via phase-contrast MRI, Doppler ultrasound, or derived from population-based waveforms scaled by cross-sectional area [113].
- **Outlet conditions** may use zero-pressure assumptions, fixed resistance models, or multi-element Windkessel circuits (0D lumped parameter models) to mimic the impedance of downstream microcirculation [114].
- **Wall behavior** is typically assumed to be rigid in basic simulations, but fluid–structure interaction (FSI) models can incorporate wall elasticity to capture dynamic vessel deformation, which is important for aneurysm or stent studies [107].

Mesh resolution (element size and quality) and time step size are selected to balance

numerical stability with computational cost. Solver choices include finite volume methods, as implemented in *OpenFOAM* [62], [116], finite element approaches, as in *SimVascular* [89], [106] or commercial packages such as *ANSYS Fluent* [4], and lattice Boltzmann methods for highly parallelizable flow simulations [41].

Derived Hemodynamic Biomarkers

CFD simulations yield velocity, pressure, and wall shear stress fields throughout the domain. From these, a wide range of clinically meaningful biomarkers can be computed:

- **Pressure-based metrics:** Including pressure drops, gradients, and fractional flow reserve (FFR), used to assess the functional severity of coronary stenoses [103].
- **Shear-based metrics:**
 - *Wall shear stress (WSS)* quantifies tangential forces exerted by blood flow on the vessel wall. Low WSS is associated with atherosclerotic plaque development, while high WSS may indicate aneurysmal wall degradation.
 - *Oscillatory shear index (OSI)* captures the directional variability of WSS over the cardiac cycle, with high OSI linked to disturbed flow and endothelial dysfunction.
 - *Relative residence time (RRT)* combines WSS magnitude and OSI to identify zones of recirculation or stagnation.
- **Velocity-based metrics:** Including peak systolic velocities, secondary flows, recirculation zones, and helical flow structures, particularly relevant in bifurcations or aneurysm sacs.

These metrics support a broad range of applications: hypothesis testing in disease progression (e.g., aneurysm rupture risk), virtual stenting and device deployment studies, and physiology-based planning for revascularization strategies [101], [106]. Tools such as CRIMSON [6], SimVascular [106], and ANSYS Fluent [4] are commonly used for modeling workflows, with growing support for patient-specific boundary condition tuning and mesh adaptation.

Clinical Integration and Validation

In clinical settings, CFD-derived biomarkers are increasingly used to supplement anatomical imaging. For example, non-invasive computation of fractional flow reserve from CT angiography (FFRCT) has been FDA-approved and shown to improve diagnostic accuracy for coronary artery disease, reducing the need for invasive catheterization [38]. Similarly, CFD models have been employed to evaluate aneurysm hemodynamics and guide endovascular coiling or stent placement.

Despite their promise, vascular simulations remain sensitive to input uncertainties—from segmentation errors and geometric artifacts to assumptions in boundary conditions. As such, validating simulated biomarkers against in vivo measurements (e.g., catheter-based pressure recordings, 4D flow MRI) remains a critical challenge in translational CFD research [95].

2.5 Disease-Focused Applications and Clinical Evidence

CFD-based simulations have shown strong clinical utility in a variety of vascular pathologies. Below I list concrete applications supported by the cited works.

- **Aneurysms, stenoses, and AVMs:**

- *End-to-end, patient-specific CFD workflows* for anatomy reconstruction, meshing, and hemodynamic analysis to support preoperative planning and research [106].
- *Comparative evaluation of endovascular strategies*, using CFD biomarkers (pressure/WSS/velocity) to rank or contrast treatment options for lesions or aneurysms (e.g., alternative device configurations or approaches) [18].
- *Treatment planning for aneurysms*, predicting post-intervention changes (coiling, stent/flow-diverter deployment) in intra-saccular flow, WSS, and recirculation metrics to guide device selection and placement [30], [34].
- *Quantifying intervention impact in complex cerebrovascular disease (including AVMs)*, relating simulated hemodynamic changes before/after therapy to

expected clinical effects or follow-up endpoints [56].

- **Coronary artery disease (CAD):**

- *Physiology-guided revascularization with invasive FFR*, using translesional pressure ratios to determine the need for Percutaneous Coronary Intervention (PCI) in intermediate stenoses; supported by randomized trial evidence and follow-up analyses [66], [67], [103].
- *Perfusion imaging as a functional gatekeeper*, employing static and dynamic myocardial perfusion measurements to identify ischemia, refine diagnosis, and reduce unnecessary invasive procedures [17], [61].
- *CT-based diagnostic pathways and detection performance*, incorporating functional information alongside anatomic assessment to improve CAD workup and decision-making [7].

2.6 Limitations of Real-World Clinical Data

Despite their potential, patient-specific simulations and learning-based models face key bottlenecks:

- **Imaging limitations:** CTA resolution often excludes vessels under $\sim 2\text{mm}$ in diameter, resulting in incomplete reconstructions [87].
- **Segmentation noise:** Semi-automated methods struggle with artifacts and anatomical variability, particularly in tortuous or branching regions [2].
- **Boundary condition uncertainty:** Simplified models like Murray’s law or generic outlet resistances can introduce errors in flow estimation [110], [117].
- **Privacy and data scarcity:** Regulatory constraints and limited labeled datasets—especially for rare diseases—hinder generalization and reproducibility [43], [75].

These factors create strong incentives to explore alternative sources of vascular data through synthesis.

2.7 Synthetic Vascular Data Generation: Model-Driven

Synthetic data generation offers a scalable, privacy-preserving route to build diverse vascular cohorts for method development, benchmarking, and data augmentation. Model-driven approaches specify rules and constraints that encode vascular physiology and geometry, producing families of anatomically plausible networks without direct access to patient data. Early strategies include:

- **Fractal-based models:** Procedural generation using asymmetry and flow–diameter relationships [29], [123].
 - *Mechanics.* Networks are grown recursively by applying branching rules (e.g., bifurcation asymmetry, diameter scaling, and branch angle distributions). A tree expands from one or multiple roots until stopping criteria are met (e.g., terminal radius, maximum order, or coverage goals).
 - *Typical parameterization.* User-defined laws for (i) diameter ratios at bifurcations, (ii) segment length and tapering, (iii) branching angles, and (iv) asymmetry indices; optional constraints enforce minimum separation or collision avoidance.
 - *Outputs and use cases.* Centerline graphs with per-segment radii that can be turned into lumen surfaces (via sweep/tube reconstructions) for meshing; useful for stress-testing segmentation/registration or training topology-aware models.
 - *Strengths.* High controllability, fast generation, and interpretable knobs that map to physiological intuitions [29], [123].
 - *Limitations.* Limited realism in complex regions (e.g., tortuous or pathological anatomy), sensitivity to hand-tuned parameters, and difficulty matching population-level statistics across multiple vascular beds [29], [123].
- **Space-filling algorithms and Constrained Constructive Optimization (CCO):** Tree-like structures are grown within perfusion territories by minimizing cost functions under hemodynamic constraints [31], [36], [55], [100].
 - *Mechanics.* Given a perfusion volume and one or more inlets, terminal sites

are iteratively placed and connected. variants add candidate branches and select the one that minimizes a global cost (e.g., total intravascular volume, pumping power) while satisfying flow–radius and path-length constraints.

- *Typical parameterization.* (i) Tissue/perfusion domain and inlet location(s); (ii) terminal density or demand; (iii) hemodynamic rules linking flow and radius; (iv) geometric constraints on angles, minimum lengths, curvature, and proximity; (v) stopping conditions (coverage or terminal count).
- *Outputs and use cases.* Large, supply-consistent trees that “fill” a target organ or territory; suitable for generating coronary, hepatic, or cerebral territories for downstream meshing, CFD boundary assignment, or algorithm stress tests [31], [36], [55], [100].
- *Strengths.* Better spatial coverage and physiological plausibility in perfusion-constrained settings; explicit coupling between geometry and flow surrogates via the optimization objective [31], [36], [55], [100].
- *Limitations.* Computational cost scales with territory size and candidate evaluation; realism depends on cost design and constraint calibration; post-processing may be required to correct sharp turns or to smooth junctions [31], [36], [55], [100].

While interpretable and controllable, model-driven approaches can lack fine anatomical realism and often require manual parameter tuning. They may fail to reproduce the full variability observed across individuals and pathologies even when rules encode hemodynamic considerations.

2.8 Synthetic Vascular Data Generation: Data-Driven

The rise of deep generative models has opened new possibilities for vascular synthesis, enabling data-driven learning of complex shape and topology distributions. Below we highlight representative approaches and their practical trade-offs.

- **Wolterink et al:** [118] (*GANs on coronary centerlines*).
 - *Representation:* Vascular trees encoded as centerline sequences; synthesis targets centerline as a point cloud rather than explicit surfaces.

- *Strengths*: Adversarial training promotes realism in global course and curvature; data-efficient compared with dense 3D supervision; well-suited for augmenting topology-aware tasks.
- *Limitations*: Does not allow branching structures; no explicit lumen surface or wall geometry; requires post-processing to enforce connectivity and physiological constraints.
- **Sinha et al:** [90] (*Diffusion on SDFs*).
 - *Representation*: Signed distance fields (SDFs) learned with diffusion, enabling resolution-independent implicit surfaces.
 - *Strengths*: Produces high-fidelity, watertight surfaces; captures smooth morphology and fine geometric detail; straightforward surface extraction (zero level set) and meshing.
 - *Limitations*: Topology is implicit—not explicitly constrained—so centerline connectivity/branching may require post hoc extraction and pruning; training cost and memory scale with volumetric fields.
- **Deo et al:** [19] (*Latent diffusion for aneurysm masks under limited data*).
 - *Representation*: Segmentation masks embedded in a learned latent space; diffusion operates in the low-dimensional latent to improve sample efficiency.
 - *Strengths*: Few-shot viability; diverse sampling of localized pathologies (aneurysmal bulges) with controllable variation; practical for data-scarce disease phenotypes.
 - *Limitations*: Focuses on local morphology (lesion masks) rather than complete vascular networks; downstream conversion to centerlines/surfaces may be needed for simulation tasks.
- **Kuipers et al:** [42] (*Point-cloud diffusion for global vasculature*).
 - *Representation*: Unordered 3D point sets representing vessel centerlines or sampled surfaces; diffusion steps act directly on point clouds.
 - *Strengths*: Captures large-scale geometry and global shape variation; flexible density control; avoids voxel memory costs.

- *Limitations:* Lack of explicit connectivity—requires graph reconstruction heuristics to form trees; mesh derivation needs additional surface fitting.
- **Prabhakar et al:** [69] (*Graph-based diffusion for vascular topology*).
 - *Representation:* Vessels as graphs (nodes/edges) with attributes (e.g., radii, positions); diffusion defined over graph structure to preserve connectivity.
 - *Strengths:* Explicit control of branching patterns, hierarchy, and cycles; naively preserves connectivity—useful for centerline analytics and network-level studies.
 - *Limitations:* Surface geometry is indirect—requires geometric lifting (e.g., radius-to-surface) to obtain meshes; fine morphological details depend on separate reconstruction stages.

Trade-offs and design choices: SDF-based diffusion emphasizes *geometric fidelity* and watertight surfaces, but needs extra steps to recover explicit topology; graph-based diffusion emphasizes *topological correctness* and connectivity, but requires geometric lifting to obtain CFD-ready surfaces. Point-cloud diffusion provides a middle ground with global shape flexibility but demands post-processing for connectivity and meshing. GANs on centerlines remain attractive for topology-aware augmentation with modest supervision, while latent diffusion on lesion masks excels in *data-limited* regimes targeting localized pathology. In practice, many pipelines combine these representations to approach clinically compatible outputs using complementary strengths of the above methods.

2.9 Artificial Neural Networks

Artificial Neural Networks are computational models inspired by the structure and functioning of the human brain. They consist of a set of interconnected units called *neurons*, organized in layers. Each neuron performs a simple mathematical operation: it receives a set of inputs, weights them using a vector of parameters, applies a nonlinear activation function, and produces an output.

Formally, if a neuron receives an input $\mathbf{x} \in \mathbb{R}^n$ and weights $\mathbf{w} \in \mathbb{R}^n$, its output y is defined as:

$$y = \phi(\mathbf{w}^\top \mathbf{x} + b)$$

where ϕ is an activation function and b is a bias term.

Two commonly used activation functions are:

- **Rectified Linear Unit (ReLU):**

$$\text{ReLU}(z) = \max(0, z)$$

This function introduces nonlinearity into the model and allows for efficient gradient propagation during training, making it especially useful in deep networks.

- **Leaky ReLU:**

$$\text{LeakyReLU}(z) = \begin{cases} z, & \text{if } z \geq 0 \\ \alpha z, & \text{if } z < 0 \end{cases}$$

where α is a small constant (e.g., 0.01). Unlike ReLU, it allows a small, non-zero gradient when $z < 0$, helping to mitigate the “dying ReLU” problem.

- **Hyperbolic Tangent (tanh):**

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

This function maps the input to a range between -1 and 1 , which can be useful when centering the data around zero.

By organizing multiple neurons into layers and stacking them, one obtains an architecture capable of approximating complex functions, enabling the solution of tasks such as classification, regression, segmentation, and data generation. Learning in a neural network consists of adjusting the weights $\mathbf{w}$ and biases b to minimize a loss function defined over a training dataset, typically using optimization algorithms such as gradient descent.

Deep Neural Networks (DNNs), which include many hidden layers, have demonstrated outstanding performance in various fields such as computer vision, natural language processing, and bioinformatics. While deep neural networks excel at learning from structured vector inputs, they are inherently limited when applied to data with non-Euclidean or hierarchical structure, such as trees or graphs. In such domains, the relationships between elements are defined by topological connections rather than fixed spatial or sequential

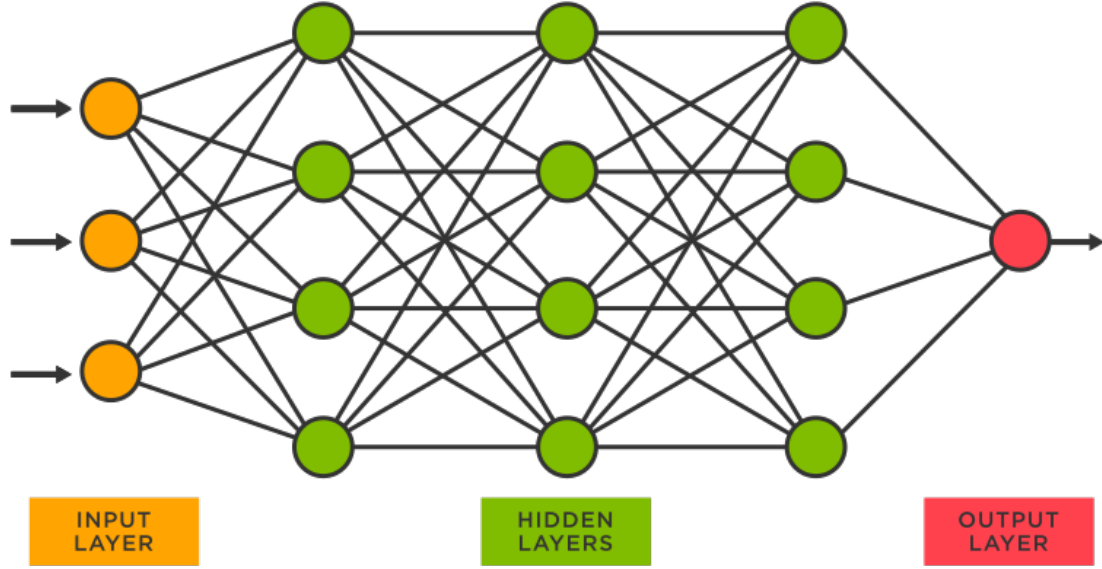


Figure 2.4: Architecture of a deep neural network composed of an input layer, multiple hidden layers, and an output layer. Each layer consists of several neurons, and each neuron applies a nonlinear activation function to its weighted input. Connections between layers are fully connected.

order. Standard architectures fail to exploit these structural dependencies.

2.10 Recursive Neural Networks

RvNNs are a class of neural architectures designed to process hierarchically structured data, such as trees or directed acyclic graphs. Unlike feedforward or recurrent neural networks, which operate on flat or sequential inputs, RvNNs explicitly exploit hierarchical topologies by recursively aggregating information from child nodes to compute latent representations at parent nodes.

Formally, a RvNN operates on a tree T composed of nodes $\{v_i\}$, where each node v_i is associated with a feature vector x_i . For a non-leaf node with child representations $\{h_j\}$, the hidden state h_i is computed using a composition function f parameterized by learnable weights θ :

$$h_i = f(h_{j_1}, h_{j_2}, \dots, h_{j_k}; \theta).$$

This process is applied recursively in a bottom-up manner, starting from the leaves and

continuing until a global representation is obtained at the root node. Such hierarchical aggregation enables the model to capture long-range structural dependencies and encode context-sensitive information in a compact form.

RvNNs have been successfully applied in a variety of domains where structured representations are essential. In natural language processing, they have been used for sentiment analysis and syntactic parsing by modeling sentence parse trees [91]. In computer graphics and shape analysis, they have enabled the reconstruction of hierarchical 3D geometries [48], where objects are decomposed into part-based tree structures.

Due to their ability to capture both local and global dependencies in hierarchical data, RvNNs provide a powerful inductive bias for learning over structured domains. In this thesis, we leverage this architecture to model the topology and geometry of vascular trees as recursive compositions of their branching segments.

2.11 Autoregressive Architectures: Transformers

Transformer architectures are a class of deep learning models that rely on self-attention mechanisms to process and generate sequences (Fig. 2.5). Unlike Recurrent Neural Networks (RNNs), which operate sequentially and maintain an internal state to capture temporal dependencies, transformers dispense with recurrence entirely. Instead, they leverage attention to allow each position in the input sequence to attend to all other positions simultaneously, enabling highly parallelizable computation and improved modeling of long-range dependencies [112].

In an autoregressive setting, the transformer models the joint distribution of a sequence $x = (x_1, x_2, \dots, x_n)$ as a product of conditional probabilities:

$$p(x) = \prod_{t=1}^n p(x_t \mid x_{<t}),$$

where each token x_t is predicted given all previous tokens. To enforce this causal structure during training, the model uses a *masked self-attention* mechanism that prevents each position from accessing future tokens.

At the core of the transformer is the self-attention mechanism, which computes pairwise interactions between tokens in the sequence. Given an input sequence represented as

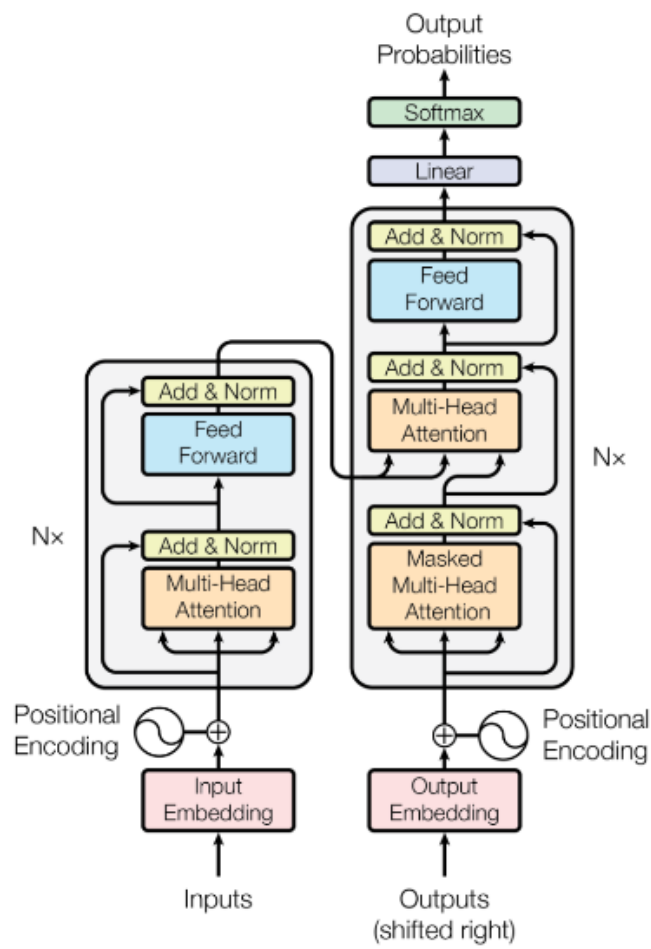


Figure 2.5: The Transformer - model architecture[112]

a matrix $X \in \mathbb{R}^{n \times d}$, the model projects it into three matrices: queries Q , keys K , and values V , each of dimension $n \times d_k$. The attention scores are then computed as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V.$$

This operation allows each token to aggregate information from the entire sequence, weighted by content similarity. To increase representational capacity, transformers use *multi-head attention*, which applies this mechanism in parallel across multiple subspaces.

A transformer decoder block comprises several key components: a masked multi-head self-attention layer, a position-wise feedforward network (typically two linear layers with a nonlinearity in between), residual connections, and layer normalization. These blocks are stacked to form deep models that can capture complex dependencies across long sequences.

To encode positional information (which transformers do not inherently model due to their non-sequential nature), positional encodings—either learned or sinusoidal—are added to the input embeddings. The result is a model that can conditionally generate each token in a sequence given its context and can be trained efficiently using teacher forcing—that is, during training, the decoder is fed the ground-truth token from the previous step instead of its own prediction—together with a cross-entropy loss.

Transformers have become the backbone of state-of-the-art models in diverse domains including natural language processing (e.g., GPT [73], BERT [20]) and image generation (e.g., DALL · E [78]).

2.12 3D Data Representations

Three-dimensional data can be represented in various formats depending on the requirements of the application, such as fidelity, compactness, and suitability for machine learning or physical simulation. Common 3D data representations include surface meshes, point clouds, Signed Distance Functions (SDFs), and volumetric grids. Each representation offers specific advantages and trade-offs in terms of accuracy, memory efficiency, and ease of manipulation.

2.12.1 Surface Meshes

Surface meshes are among the most widely used representations in biomedical modeling. They consist of a set of *vertices* connected by *edges* to form *polygonal faces*, typically triangles or quadrilaterals, that define the outer surface of a structure (Fig. 2.6). This makes them particularly suitable for representing anatomical boundaries such as vascular walls, organ surfaces, or bone structures.

Meshes are:

- **Compact:** Unlike volumetric representations, they store only the surface geometry, leading to lower memory usage.
- **Highly expressive:** Capable of accurately capturing complex shapes with fine-grained detail, depending on the resolution.
- **Versatile:** Widely supported across simulation, visualization, and 3D printing pipelines.

In biomedical contexts, meshes play a central role in several key applications:

- **Finite Element Methods (FEM):** Used for biomechanical simulations such as stress analysis on bones or vascular walls.
- **Computational Fluid Dynamics (CFD):** Essential for simulating blood flow in arterial or venous systems, often using patient-specific mesh reconstructions.
- **Morphometric analysis:** Enabling shape comparison, anatomical atlases, and statistical modeling of population-level variability.

However, surface meshes also present several challenges:

- **Irregular Topologies:** Biomedical structures often exhibit non-manifold edges, which can complicate mesh generation and processing.
- **Mesh Quality:** Poorly conditioned elements (e.g., skinny triangles, inconsistent normals) can negatively affect the stability and accuracy of numerical solvers.
- **Incompatibility with standard deep learning and training challenges:** Traditional convolutional architectures expect regular grid inputs, so meshes require *graph neural networks*, *geometric deep learning*, and *mesh-based neural architectures*. Moreover, training such models poses practical difficulties, chiefly a very high

computational cost.

Despite their challenges, surface meshes remain one of the most powerful and flexible representations for 3D anatomical data, supporting a wide range of downstream applications in medical simulation, machine learning, and digital anatomy.

2.12.2 Point Clouds

Point clouds represent 3D geometry as a collection of discrete points in space, typically sampled from an object’s surface (Fig. 2.6). This format is often the raw output of 3D scanning devices such as LiDAR, structured light scanners, or photogrammetry pipelines, as well as the result of surface sampling in segmentation or meshing processes. Each point encodes spatial coordinates, and optionally, additional attributes such as color, intensity, normal vectors, or confidence scores.

Point clouds are:

- **Simple and lightweight:** Their unstructured nature makes them memory-efficient and easy to acquire.
- **Detail-preserving:** They can capture high-frequency geometric details and fine surface textures when densely sampled.
- **Format-agnostic:** They can be extracted from surface meshes, volumes, or implicit functions, making them a flexible intermediate representation.

Despite their advantages, point clouds have several inherent limitations:

- **Lack of connectivity:** They do not contain information about how points are connected, which complicates tasks like surface reconstruction, topological analysis, or physical simulation.
- **Non-uniform sampling:** Real-world point clouds often suffer from noise, missing regions, and spatially varying density.

To overcome these issues, several neural network models have been proposed for directly learning from point sets. Notably:

- **PointNet [71]:** Introduced a permutation-invariant architecture using symmetric functions (e.g., max pooling) to aggregate point features.

- **PointNet++** [72]: Extended PointNet by introducing hierarchical feature learning through spatial grouping and local neighborhoods.
- **DGCNN** [115]: Modeled point cloud data as dynamic graphs, enabling edge-based convolutions that better capture local geometry.

Point clouds are also commonly used as an intermediate or target representation in 3D generation, registration, and alignment tasks.

In summary, point clouds offer a flexible and efficient representation for 3D data, particularly suitable for deep learning pipelines and real-time acquisition, but their unstructured nature requires careful algorithmic handling for tasks involving topology or physical modeling.

2.12.3 Volumetric Grids and Voxels

Volumetric representations discretize 3D space into a regular grid of small cubes called *voxels* (volumetric pixels), each of which stores information about the enclosed volume (Fig. 2.6). Voxels are the 3D analog of pixels in 2D images and form the foundational data structure for many medical imaging modalities such as CT (Computed Tomography) and Magnetic Resonance Imaging (MRI) (Magnetic Resonance Imaging), where anatomy is naturally acquired in regularly spaced slices.

Each voxel may store:

- **Binary occupancy values**, indicating whether a given region of space is inside or outside a structure.
- **Scalar intensities**, such as Hounsfield units in CT or T1/T2 signal intensities in MRI.
- **Semantic or probabilistic labels**, for use in segmentation or classification tasks.

Volumetric data offer several advantages:

- **Regular grid structure:** Allows straightforward application of 3D convolutional neural networks (3D CNNs), extending classical 2D architectures to operate on volumetric data.
- **Compatibility with imaging data:** Well-matched to the native format of

most medical scans, enabling end-to-end deep learning pipelines without requiring geometric conversion.

- **Topological completeness:** Volumetric grids can represent enclosed structures and cavities with full spatial context.

However, voxel-based representations come with significant challenges:

- **Memory inefficiency:** Storage and computation scale cubically with resolution ($N \times N \times N$), making high-resolution grids computationally prohibitive.
- **Quantization artifacts:** The discrete nature of voxels can introduce blocky representations and loss of fine geometric detail, especially at lower resolutions.
- **Limited geometric fidelity:** Smooth surfaces and sharp edges are difficult to capture without very dense grids or post-processing.

To mitigate these limitations, several hierarchical or adaptive representations have been proposed:

- **Sparse voxel grids:** Store only the non-empty (active) voxels to reduce memory usage and computational cost.
- **Octrees:** Hierarchical data structures that recursively divide space into octants, enabling multi-resolution representation of complex geometries [79].
- **Hybrid methods:** Combine voxel grids with meshes, point clouds, or implicit functions for efficient geometry encoding and neural processing.

Volumetric grids provide a powerful and structured way to encode 3D data, especially for tasks aligned with medical imaging. Despite their limitations in resolution and memory footprint, their integration with 3D CNNs and compatibility with scan-based modalities makes them a fundamental representation in both traditional image analysis and modern deep learning workflows.

2.12.4 Signed Distance Functions (SDFs)

Signed Distance Functions (SDFs) represent 3D shapes implicitly by defining a continuous scalar field over space. For any point $\mathbf{x} \in \mathbb{R}^3$, the SDF assigns a scalar value $f(\mathbf{x})$ corresponding to the shortest distance from $\mathbf{x}$ to the closest point on the surface of the

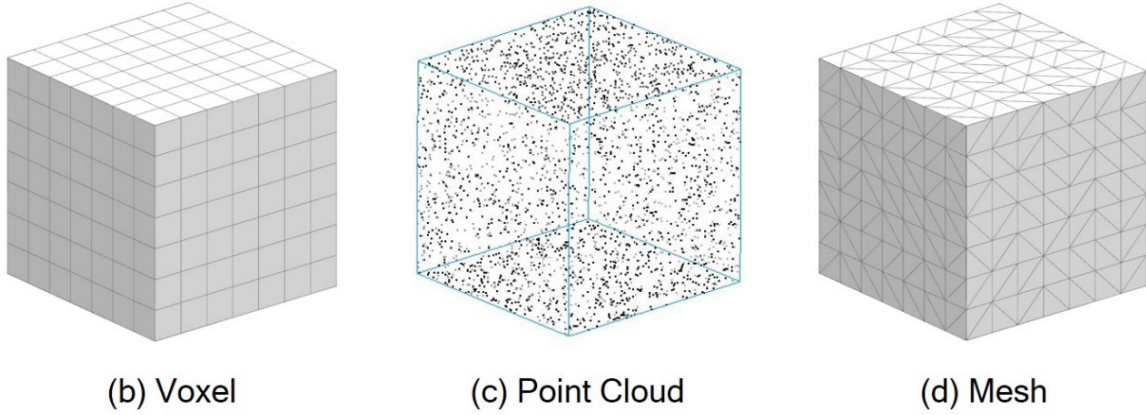


Figure 2.6: Voxel, point cloud, and polygon mesh representation of 3D models [16]

shape. The sign of this value encodes location relative to the surface: it is **negative** inside the shape, **positive** outside, and **zero** exactly on the surface boundary (Fig. 2.7).

This representation offers several powerful properties:

- **Continuity and smoothness:** Unlike discrete voxel grids or point clouds, SDFs are continuous in space and differentiable almost everywhere, making them ideal for gradient-based optimization.
- **Implicit encoding:** Surfaces are defined implicitly as the zero-level set of the SDF (i.e., $\{\mathbf{x} \mid f(\mathbf{x}) = 0\}$), allowing for watertight, manifold geometry.
- **Geometric operations:** Boolean operations (union, intersection, subtraction), morphing, and deformation can be performed directly on the scalar field.
- **Resolution independence:** SDFs can be evaluated at any spatial resolution, enabling smooth surface reconstruction at arbitrary detail.

SDFs have gained significant popularity in the context of neural implicit representations. One of the most influential methods, **DeepSDF** [63], uses a multilayer perceptron (MLP) to regress the signed distance value at any 3D coordinate, conditioned on a latent shape embedding. This formulation enables:

- Learning a continuous and compact representation of a shape family.
- Encoding and decoding arbitrary geometries from low-dimensional latent vectors.

- Performing shape interpolation, completion, and deformation in latent space.

In biomedical imaging, SDFs are particularly appealing for modeling anatomical surfaces with subvoxel accuracy and for learning priors over deformable structures such as organs or vessels. Their ability to produce watertight surfaces is especially useful in simulations or when converting implicit surfaces into triangle meshes using algorithms such as Marching Cubes.

Despite their many advantages, SDFs also come with several practical limitations:

- **High computational cost:** Evaluating SDFs can be computationally expensive, as each query point requires a forward pass through a neural network. Dense voxel-based SDFs also consume substantial memory at high resolutions.
- **Implicit geometry representation:** Since the surface is not explicitly defined, tasks such as rendering, collision detection, or simulation require extracting the zero level set (e.g., via Marching Cubes), which adds complexity and introduces discretization artifacts.
- **Difficulty with fine structures:** Capturing thin or highly detailed anatomical features (e.g., narrow vessels or bifurcations) can be challenging, particularly when training data is sparse or noisy.
- **Numerical instability near surfaces:** The gradient of the SDF can become unstable near the zero level set, leading to imprecise surface reconstruction or flat regions when extracting meshes.
- **Lack of explicit topology:** SDFs do not encode topological structure directly, which can be a disadvantage in applications like vascular modeling where structural constraints are important.

In summary, SDFs provide a powerful and flexible implicit representation of 3D shapes. Their differentiability and latent encoding capabilities make them especially suited for deep generative modeling, physical simulation, and shape optimization in medical and graphics applications. However, their implicit nature and computational cost pose challenges for tasks that require direct surface access, fine structural detail, or real-time performance. Addressing these limitations remains an active area of research, particularly in the context of high-fidelity biomedical modeling.

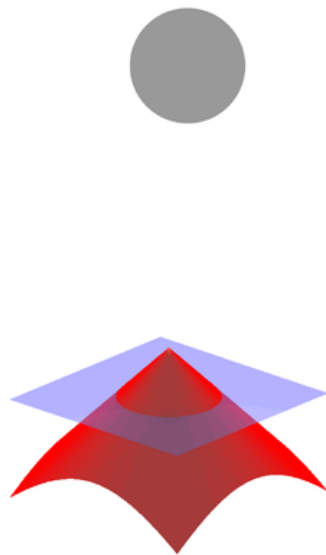


Figure 2.7: Illustration of a signed distance field (SDF): the gray circle is the original shape, the red surface shows the distance to that shape, and the blue plane is a 2D slice where this distance is sampled. Source: Oleg Alexandrov, Public domain, via Wikimedia Commons. Link: https://commons.wikimedia.org/wiki/File:Signed_distance1.png

2.12.5 Hybrid and Parametric Representations

Hybrid representations seek to combine the strengths of multiple formats to better capture the complexity of anatomical structures. For example, point clouds may be used to represent fine-grained surface geometry, while centerlines or graphs encode the underlying topology of branching structures such as blood vessels or airways. Similarly, volumetric data can be enriched with mesh-based surface priors or augmented with implicit fields for smoother interpolation and completion.

Parametric models offer an alternative approach by encoding geometry using compact, mathematically defined primitives. B-spline curves, tubular templates, or medial axis representations are commonly employed in vascular modeling, where the structure adheres to anatomical constraints. These models facilitate operations such as morphing, interpolation, and statistical shape analysis, and are especially useful for population-level studies, atlas construction, or individualized geometry fitting. Their regularity and low-dimensional control parameters make them efficient and interpretable, albeit at the cost of reduced flexibility in capturing complex or pathological morphologies.

2.12.6 Choice of Representation

The optimal choice of 3D representation depends on the intended application, the nature of the data, and computational considerations. Surface meshes are preferred for applications requiring watertight geometry, such as CFD or finite element methods (FEM). Point clouds offer simplicity and memory efficiency, making them suitable for acquisition pipelines and lightweight deep learning models. Voxel grids are directly compatible with imaging modalities like CT and MRI, and benefit from mature convolutional architectures, though their memory demands can be prohibitive at high resolution.

Implicit representations such as signed distance functions (SDFs) and occupancy fields provide continuity and differentiability, making them attractive for generative modeling and shape optimization. However, their implicit nature can complicate surface extraction and real-time interaction. Parametric models, while highly efficient for specific anatomical domains, may lack the expressiveness needed for modeling arbitrary or abnormal geometries.

In recent years, deep learning methods have begun to exploit hybrid and learned representations that combine the structural regularity of meshes, the flexibility of point clouds,

and the continuity of implicit fields. The growing ecosystem of geometric deep learning tools has expanded the design space for 3D learning, enabling task-specific combinations tailored to reconstruction, simulation, segmentation, or generation.

Ultimately, the choice of representation involves trade-offs between fidelity, interpretability, computational cost, and suitability for learning or simulation. Understanding these trade-offs is essential when designing pipelines for biomedical modeling, particularly in settings where clinical precision, scalability, and generalizability are critical.

Chapter 3

VesselVAE: Recursive Variational Autoencoders for 3D Blood Vessel Synthesis

We first present a framework named VesselVAE for synthesizing blood vessel geometry. This generative framework is based on a Recursive variational Neural Network (RvNN), that has been successfully applied in various contexts, including natural language [91], [92], shape semantics modeling [48], [49], and document layout generation [64]. In contrast to previous data-driven methods that generate vascular structures, our recursive network fully exploits the hierarchical organization of the vessel and learns a low-dimensional manifold encoding branch connectivity along with geometry features describing the target surface. Once trained, the VesselVAE latent space is sampled to generate new vessel geometries. We trained the model on different datasets of cerebral arteries, including aneurysms. We perform extensive experiments with our synthetically generated datasets to assess the ability of our method to generate high quality vessel geometry.

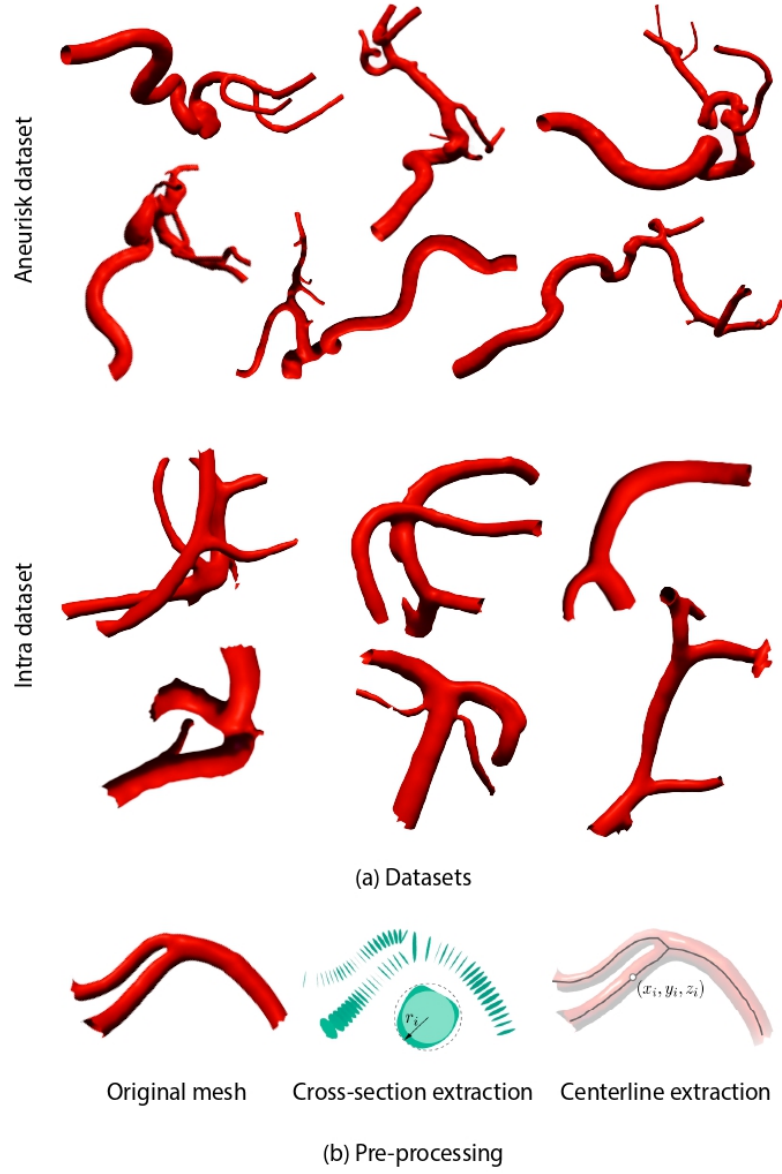


Figure 3.1: Datasets and pre-processing overview: (a) The raw meshes from the Intra 3D collection (b) The meshes undergo pre-processing using the VMTK toolkit. This step is crucial for extracting centerlines and cross-sections from the meshes, which are then used to construct their binary tree representations.

3.1 Methods

3.1.1 Input

The network’s input is a binary tree representation of the blood vessel 3D geometry. Formally, each tree is defined as a tuple $(T, \mathcal{E})$, where T is the set of nodes, and $\mathcal{E}$ is the set of directed edges connecting a pair of nodes (n, m) , with $n, m \in T$. In order to encode a 3D model into this representation, vessel segments are parameterized by a central axis consisting of ordered points in Euclidean space, each point has coordinates x, y, z and radius r , assuming a piece-wise tubular vessel for simplicity. We then construct the binary tree as a set of nodes $T = n_1, n_2, \dots, n_N$, where each node n_i represents a vessel segment and contains an attribute vector $\mathbf{x}_i = [x_i, y_i, z_i, r_i] \in \mathbb{R}^4$ with the coordinates of the corresponding point and its radius r_i . See Sec. 3.2 and Fig. 3.2 for details.

3.1.2 Network architecture

The proposed generative model is a Recursive variational Neural Network (RvNN) trained to learn a probability distribution over a latent space that can be used to generate new blood vessel segments (Fig. 3.2). The framework follows an encoder-decoder structure, and the input is a binary tree representation of the vessel $\mathbf{x}$. During training, the encoder (*Enc*) maps this tree to a compact Gaussian distribution in the learned latent space, ensuring consistency with a standard normal distribution across all training exemplars. After encoding, the decoder (*Dec*) draws samples from a multivariate Gaussian distribution: $\mathbf{z}_s(\mathbf{x}) \sim N(\mu, \sigma)$, with $\mu = f_\mu(\text{Enc}(\mathbf{x}))$ and $\sigma = f_\sigma(\text{Enc}(\mathbf{x}))$, and learns to reconstruct the input vessel anatomies. After training, the RvNN can generate new geometry by drawing samples from a standard normal distribution and decoding them into novel blood vessel trees $\mathbf{x}'$.

Encoder network

The encoding process begins with a depth-first post order traversal of the tree $\mathbf{x}$, recursively encoding each node together with its child nodes, beginning from the leaf nodes until the root node is reached (Fig. 3.3, (a)). First, the node’s attribute vector is processed to produce a partial encoding. If the node has two children, their attribute vectors—along with those of any descendants—are already recursively encoded. The network subsequently

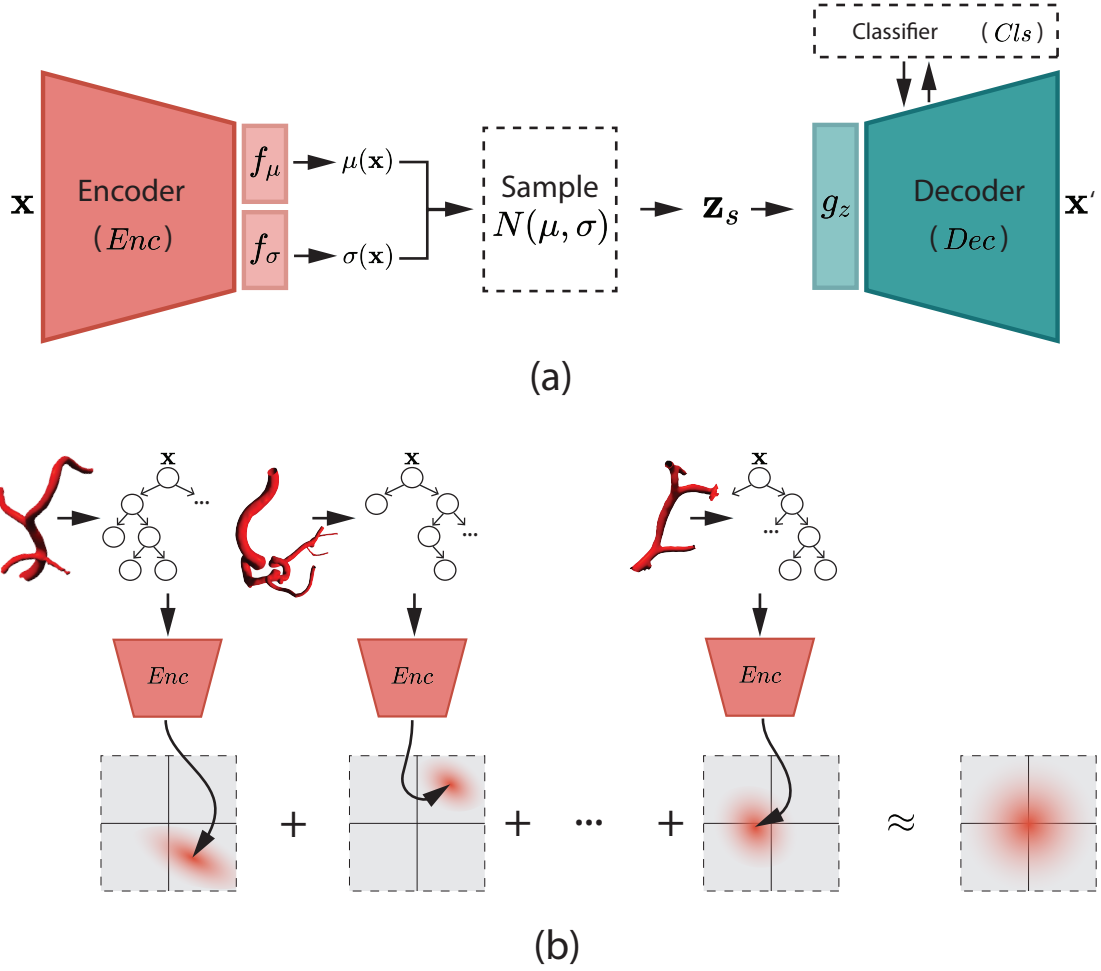


Figure 3.2: (a) Overview of the Recursive variational Neural Network for synthesizing blood vessel structures. The architecture follows an Encoder-Decoder framework which can handle the hierarchical tree representation of the vessels. The encoder network (Enc) maps a training exemplar to a compact Gaussian distribution in the latent space, while the generator maps a sample from this Gaussian back to the original exemplar. (b) The combination of per-exemplar Gaussians is restricted to match the standard normal distribution (bottom). In the testing phase, drawing samples from the standard normal and decoding them through the generator allows for exploration of the probabilistic manifold captured from the training set. Note that vessel geometry is represented as a binary tree ($\mathbf{x}$) before being passed through the encoder.

processes the latent vectors obtained from the right and left child nodes through additional layers, sums the resulting representations, and then concatenates them with the node's

partial encoding. If the node has only one child, the same procedure is followed using a single latent vector. For leaf nodes, a zero vector of the appropriate dimension is concatenated in place of the missing child branches. Ultimately, every node produces a D dimensional encoding that captures the attributes of both the node and its entire sub-tree. Repeating this process up the hierarchy yields a single latent vector representing the entire tree. See Fig. 3.3, a more detailed figure illustrating the encoder’s internal structure. In addition to the core architecture, our Encoder is further augmented with two auxiliary shallow fully-connected neural networks: f_μ and f_σ . Positioned before the RvNN sampling step, these shallow networks shape the distribution of the latent space where the encoded tree structures lie.

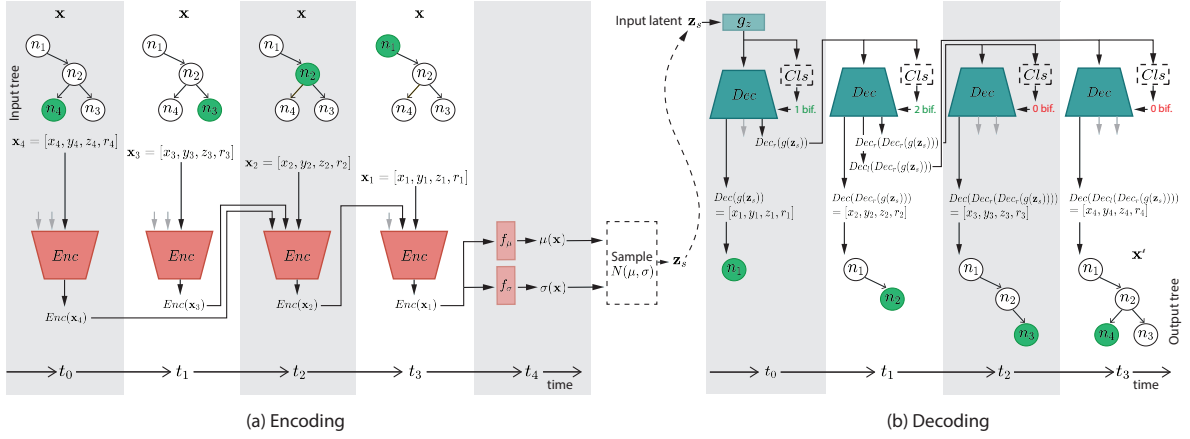


Figure 3.3: Proposed encoder-decoder framework. (a) Encoding. The input vessel tree is traversed over time steps ($t_0 \rightarrow t_3$, encoding side) from leaves to root. For each node (indicated in green), a feature vector $\mathbf{x}_i = [x_i, y_i, z_i, r_i]$ is processed by the encoder Enc to produce latent embeddings that are ultimately merged into $\mu(\mathbf{x})$ and $\sigma(\mathbf{x})$ parameters of a Gaussian distribution in t_4 . (b) Decoding. A sample from this distribution is then fed through the decoder Dec to hierarchically reconstruct the tree ($t_0 \rightarrow t_3$, decoding side). At each step, the model reconstructs output features for the node $\mathbf{x}_i = [x_i, y_i, z_i, r_i]$ and either continues branching or terminates (via the Cls outputs), eventually reproducing the full vessel-like structure. Green circles highlight nodes created in each step.

Decoder network

The decoding phase reverses the encoding process, reconstructing a binary tree representation $\mathbf{x}'$ from the sampled latent vector $\mathbf{z}_s$, as depicted in (Fig. 3.3, (b)). The first

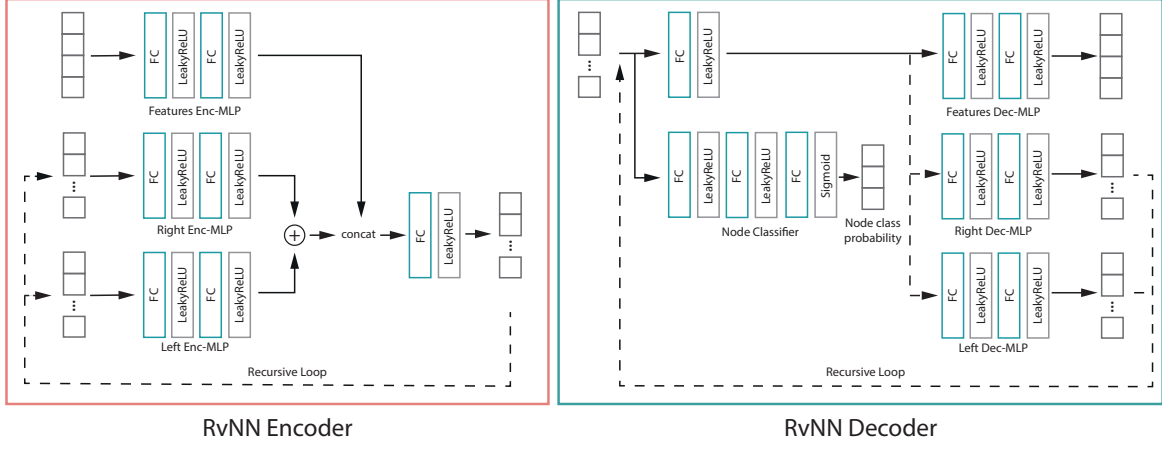


Figure 3.4: Layers of the Encoder and Decoder networks comprising branches of fully-connected layers followed by leaky ReLU activations. Notice that right/left Enc-MLPs of the Encoder only execute when the incoming tree requires it. Similarly, the Decoder only uses right/left Dec-MLPs when the Node Classifier predicts bifurcations.

step is to decode the root node. Then, on each recursive step, the network decides how many child nodes to decode (one, two, or none). We equipped the decoder with a node classifier module Cl_s , which predicts whether a given latent code corresponds to a leaf node or to an internal node with one or two bifurcations. This is implemented as a multi-layer perceptron trained to predict a three-category bifurcation probability. Complementing the node classifier, the decoder layers are responsible for reconstructing the attribute vectors for each node, specifically its coordinates and radius. Additionally, the decoder can generate up to two extra latent codes (left and right) to enable the recursive decoding of the next node in the hierarchy. If the classifier predicts that only one child is present, we assign it as the right child by default. The Decoder is further augmented with an auxiliary shallow fully-connected neural network: g_z situated after the sampling to facilitate the decoding of latent variables, aiding the Decoder network in the reconstruction of tree structures. Collectively, this supplementary network together with the ones of the Encoder (f_μ and f_σ) streamline the data transformation process through the model. All activation functions used in our networks are leaky ReLUs, except for g_z which uses the Tanh. A complete description of the networks, including a table detailing layers, activation functions, and parameter counts, is provided in Tab. 3.1 and Fig. 3.4.

Layer name	Input shape	Output shape	Layer name	Input shape	Output shape
Encoder			Decoder		
fc1	4	512	fc1	64	256
fc2	512	64	fc_left1	256	256
right_fc1	64	512	fc_left2	256	64
right_fc2	512	64	fc_right1	256	256
left_fc1	64	512	fc_right2	256	64
left_fc2	512	64	fc2	256	64
fc3	128	64	fc3	64	4
Supplementary Networks					
$f_\mu + f_\sigma$			g_z		
fc1	128	512	fc1	64	256
fc2mu	512	128	fc2	256	256
fc2var	512	128	fc3	256	64
			Classifier		
-	-	-	fc1	64	256
-	-	-	fc2	256	256
-	-	-	fc3	256	3

Table 3.1: Our proposed method consists of two primary components: a recursive Encoder and a recursive Decoder. All layers in each component are fully connected with leaky rectified linear unit (Leaky ReLU) activations, except for g_z which uses the Tanh. The Classifier is a sub-network inside the Decoder that predicts a label for each node in the input graph.

3.1.3 Objective

The encoder-decoder network and the node classifier are trained simultaneously in order to reconstruct the feature vectors $\mathbf{x}_i$ for each node along with the tree topology. For this task, we propose a combined objective function consisting of L_{recon} , which ensures feature vector reconstruction:

$$L_{recon} = \|Dec(g(\mathbf{z}_s(\mathbf{x}))) - \mathbf{x}\|_2. \quad (3.1)$$

Note that $\mathbf{z}_s(\mathbf{x})$ denotes a sample drawn from the latent Gaussian distribution whose parameters are computed from the input $\mathbf{x}$ as described before. We also define a loss term to enforce topological correctness using a three-class cross-entropy loss:

$$L_{topo} = - \sum_{c=1}^3 w_c x_c \log(p_c). \quad (3.2)$$

For each node, $p_c = Cls(Dec(g_z(\mathbf{z}_s(\mathbf{x}))))_c$ denotes the predicted probability that the node belongs to class c . These probabilities are the output of the classifier, which operates on the features produced by the decoder. The classes represent different topological types of nodes: *leaf nodes* (no children), *internal nodes* (one child), and *bifurcation nodes* (two children). The ground truth label x_c is a one-hot encoded vector that equals 1 if the true class of the node is c , and 0 otherwise. To address class imbalance, each class is weighted by w_c , the inverse of its frequency in the dataset.

Additionally, in line with the general framework proposed by β -VAE [33], we incorporated a Kullback-Leibler (KL) divergence term encouraging the distribution $p(\mathbf{z}_s(\mathbf{x}))$ over all training samples $\mathbf{x}$ to move closer to the prior of the standard normal distribution $p(\mathbf{z})$. This regularization term is defined as:

$$L_{KL} = D_{KL}(p(\mathbf{z}_s(\mathbf{x})) \| p(\mathbf{z})). \quad (3.3)$$

Finally, during training we minimize the following equation:

$$L = (1 - \alpha)L_{recon} + \alpha L_{topo} + \gamma L_{KL}. \quad (3.4)$$

3.1.4 3D mesh synthesis

Several algorithms have been proposed to generate a mesh representation of a 3D surface from a tree-structured centerline [119]. We first converted the centerline to a Signed Distance Function (SDF) defined over the 3D space. We discretize the SDF by computing the shortest voxel-to-centerline distance. The SDF at a voxel position $\mathbf{v} \in \mathbb{R}^3$ is defined as:

$$SDF(\mathbf{v}) = \min_{\mathbf{c} \in C} \left(\|\mathbf{v} - \mathbf{c}\|^2 - r(\mathbf{c})^2 \right), \quad (3.5)$$

where C is the set of points along the centerline, $\mathbf{c}$ is a point on the centerline and $r(\mathbf{c})$ is the radius associated with $\mathbf{c}$. This equation computes the shortest distance from the voxel position $\mathbf{v}$ to the centerline C , adjusted by radius $r(\mathbf{c})$. We then apply the Marching Cubes algorithm [54], a widely used technique in computer graphics and computational geometry to convert volumetric data into triangle meshes. This process enabled us to preserve fine anatomical details and spatial relationships within the arterial network and obtain a representation suitable for further analysis. Both the SDF generation and marching cubes algorithm were implemented using scripts from the VMTK toolkit ¹

3.2 Experimental Setup

3.2.1 Materials

We trained our networks using a subset of the open-access Intra dataset ² published by Yang et al. in 2020 [121]. This subset consisted of 1694 healthy vessel segments reconstructed from 2D MRA images of patients.

As a pre-process, we converted 3D meshes into a binary tree representation and used the *network extraction* script from the VMTK toolkit ³ to extract the centerline coordinates of each vessel model. The centerline points were determined based on the ratio between the sphere step and the local maximum radius, which was computed using the advancement ratio specified by the user. The radius of the blood vessel conduit at each centerline sample was determined using the computed cross-sections assuming a circular shape

¹<http://www.vmtk.org/documentation/vmtkscripts.html>

²<https://github.com/intra3d2019/Intra>

³<http://www.vmtk.org/vmtkscripts/vmtknetworkextraction>

(see Fig. 3.1). To improve computational efficiency during recursive tree traversal, we implemented an algorithm that balances each tree by identifying a new root. The new root guarantees the lowest tree height possible without altering the represented vessel topology. We additionally trimmed trees to a height of ten in our experiments. This decision reflects a balance between the computational demands of depth-first tree traversal in each training step and the complexity of the training meshes. We excluded from our study non binary trees with or without loops. However, non-binary trees can be converted into binary trees and it is possible to train with deeper trees at the expense of higher computational costs. Ultimately, we were able to obtain 700 binary trees from the original meshes using this approach.

3.2.2 Classifier training strategy

To calculate the L_{recon} loss term, we bypassed the classifier’s predictions and used the original tree topology for decoding during training. This approach allows us to compare node attributes effectively, even if the classifier fails to make accurate topology predictions while being trained to predict the degree of each node. Given the imbalance in node distribution across classes, we adjust the weights in the cross-entropy loss calculation by using the inverse of each class count. During preliminary experiments, we observed that accurately classifying nodes closer to the tree root is critical. This is because a miss-classification of top nodes has a cascading effect on all subsequent nodes in the tree (i.e. skip reconstructing a branch). To account for this, we introduce a weighting scheme that for each node, assigns a weight to the cross-entropy loss based on the number of total child nodes. The weight is normalized by the total number of nodes in the tree.

3.2.3 Implementation details

We developed the data preprocessing pipeline and network code using Python, VMTK and the PyTorch Framework. For centerline extraction, we set the VMTK script’s advancement ratio to 1.05. This produces multiple cross-sections at bifurcations, where we select the one with the smallest radius to ensure alignment with the principal centerline direction. All tree attributes were normalized to the range $[0, 1]$. In our training setup, the batch size was set to 4, and we used the ADAM optimizer with settings $\beta_1 = 0.9$,

$\beta_2 = 0.999$, and a learning rate of 1×10^{-4} . For Eq. 3.4, we used $\alpha = 0.3$ and $\gamma = 0.001$.

To enhance computation speed, we implemented dynamic batching [53], which groups together operations involving input trees of dissimilar shapes and different nodes within a single input graph. However, the memory footprint during training is very small ($\leq 1\text{GB}$) due to the use of a lightweight tree representation. It takes approximately 12 hours to train our models on a workstation equipped with an NVIDIA A100 GPU, 80GB VRAM, and 256GB RAM.

3.2.4 Metrics

We defined a set of metrics to evaluate our trained network’s performance. By using these metrics, we can determine how well the generated 3D models of blood vessels match the original dataset distribution, as well as the diversity of the generated output. The chosen metrics have been widely used in the field of blood vessel 3D modeling, and have shown to provide reliable and accurate quantification of blood vessels main characteristics [13], [44]. We analyzed tortuosity per branch, the vessel centerline total length, and the average radius of the tree. Tortuosity distance metric [14] is a widely used metric in the field of blood vessel analysis, mainly because of its clinical importance. It measures the amount of twistiness in each branch of the vessel. Vessel’s total length and average radius were used in previous work to distinguish healthy vasculature from cancerous malformations. Finally, in order to measure the distance across distributions for each metric, we compute the cosine similarity.

3.3 Results

We conducted both quantitative and qualitative analyses to evaluate the model’s performance. For the quantitative analyses, we implemented a set of metrics commonly used for characterizing blood vessels. We computed histograms of the radius, total length, and tortuosity for the real blood vessel set and the generated set (700 samples) in Fig. 3.5 (a). The distributions are aligned and consistent. We measured the closeness of histograms with the cosine similarity by projecting the distribution into a vector of n -dimensional space (n is the number of bins in the histogram). Since our points are positive, the results range from 0 to 1. We obtain a radius cosine similarity of .97, a total length cosine similarity of .95, and a tortuosity cosine similarity of .96. Results show high similarities

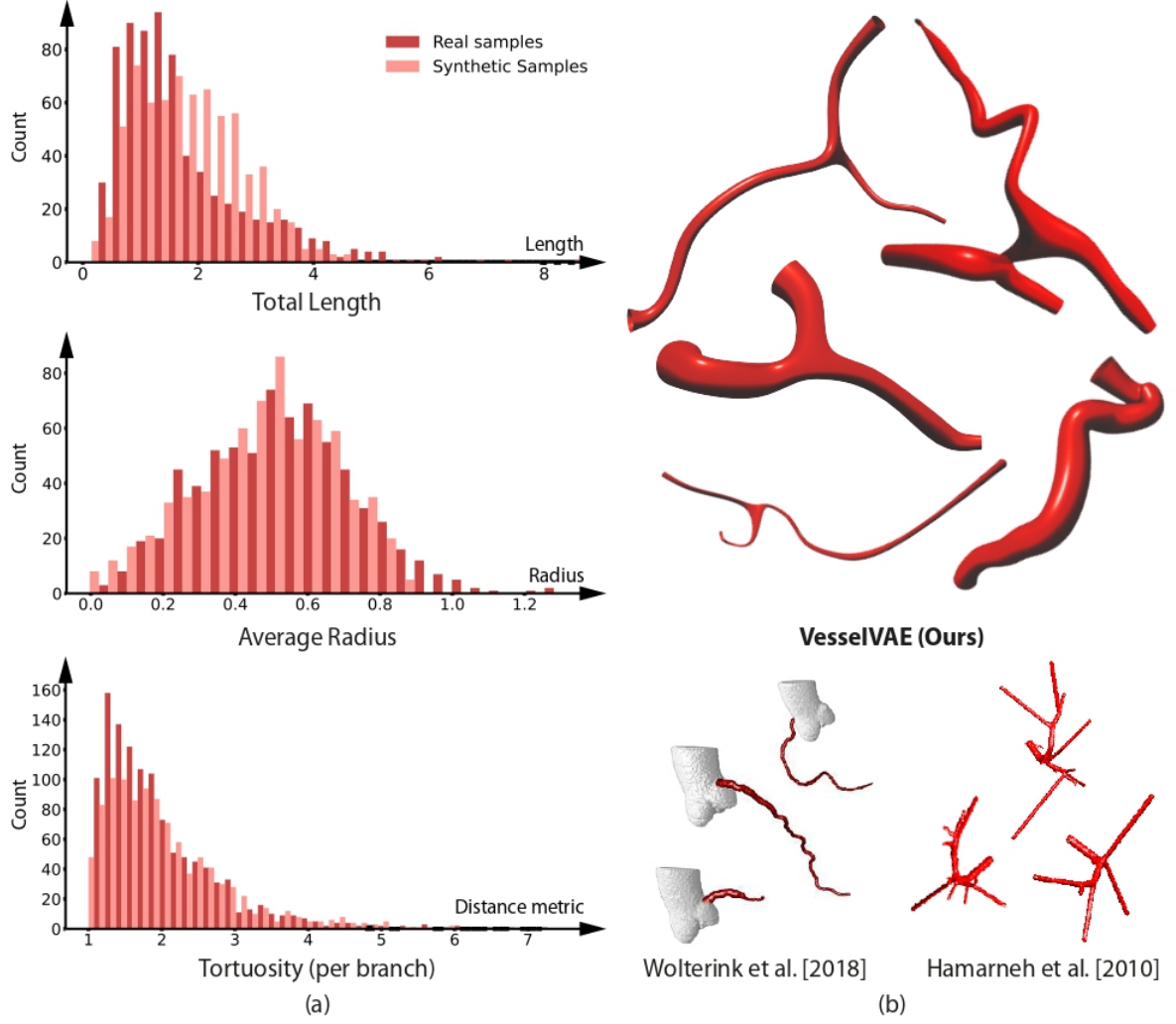


Figure 3.5: (a) shows the histograms of total length, average radius and tortuosity per branch for both, real and synthetic samples. (b) shows a visual comparison among our method and two baselines [31], [118].

between histograms demonstrating that generated blood vessels are realistic. Given the differences with the baselines generated topologies, for a fair comparison, we limited our evaluation to a visual inspection of the meshes.

The qualitative analyses consisted of a visual evaluation of the reconstructed outputs provided by the decoder network. We visually compared them to state-of-the-art methods in Fig. 3.5 (b). The method described by Wolterink and colleagues [118] is able to generate realistic blood vessels but without branches, and the method described by Hamarneh et al. [31] is capable of generating branches with straight shapes, missing on realistic modeling. In contrast, our method is capable of generating realistic blood vessels containing branches, with smooth varying radius, lengths, and tortuosity.

Chapter 4

Recursive Variational Autoencoders for 3D Blood Vessel Synthesis: Extensions, Ablations, and Broader Evaluation

In Chapter 3, we presented a framework named VesselVAE for synthesizing blood vessel geometry. This generative framework is based on a RvNN, that has been successfully applied in various contexts, including natural language [91], [92], shape semantics modeling [48], [49], and document layout generation [64]. In contrast to previous data-driven methods, our recursive network fully exploits the hierarchical organization of the vessel and learns a low-dimensional manifold encoding branch connectivity along with geometry features describing the target surface. Once trained, the VesselVAE latent space is sampled to generate new vessel geometries. In this chapter we further expand our results and analysis. Specifically, we trained on different datasets of cerebral arteries, including aneurysms. In this augmented version we performed extensive experiments with our synthetically generated datasets to assess the ability of our method to generate high quality vessel geometry. These experiments build upon and enhance our previous research by providing more comprehensive results, thereby offering a deeper understanding of the capabilities of RvNNs in the generation of anatomically plausible vascular structures.

4.1 Experimental Setup

4.1.1 Materials

We trained our networks on two different datasets. The first is a subset of the open-access Intra dataset ¹ published by Yang et al. in 2020 [121]. This subset consisted of 100 healthy vessel segments reconstructed from 2D MRA images of patients. We also trained our architecture with the open-access Aneurisk dataset ² published by the Aneurisk team [3]. This dataset is composed of 100 vessel segments reconstructed from 3D angiographic images containing healthy vessel segments and aneurysms. See Tab. 4.1 for details.

As a pre-process, we converted 3D meshes into a binary tree representation and used the *network extraction* script from the VMTK toolkit ³ to extract the centerline coordinates of each vessel model. The centerline points were determined based on the ratio between the sphere step and the local maximum radius, which was computed using the advancement ratio specified by the user. The radius of the blood vessel conduit at each centerline sample was determined using the computed cross-sections assuming a circular shape. To improve computational efficiency during recursive tree traversal, we implemented an algorithm that balances each tree by identifying a new root. The new root guarantees the lowest tree height possible without altering the represented vessel topology. We experimented with trimming the trees to different heights and analyzed the network performance for each height. The decision to use lower tree heights reflects a balance between the computational demands of depth-first tree traversal in each training step and the complexity of the training meshes (see Fig. 4.1). Note that it is still possible to train with deeper trees at the expense of higher computational costs.

Non-binary trees were transformed into binary trees through an iterative algorithm designed to split multifurcations into binary bifurcations. For each node with more than two children, the algorithm introduces intermediate nodes with the same position and features as the original. This intermediate nodes are then used to sequentially divide the children into pairs, ensuring that no node has more than two children. The process carefully preserves the original spatial arrangement and attributes of the vessel, altering only the tree topology. We excluded vessel models with loops from our experiments.

¹<https://github.com/intra3d2019/Intra>

²<https://github.com/permfl/AneuriskData>

³<http://www.vmtk.org/vmtkscripts/vmtknetworkextraction>

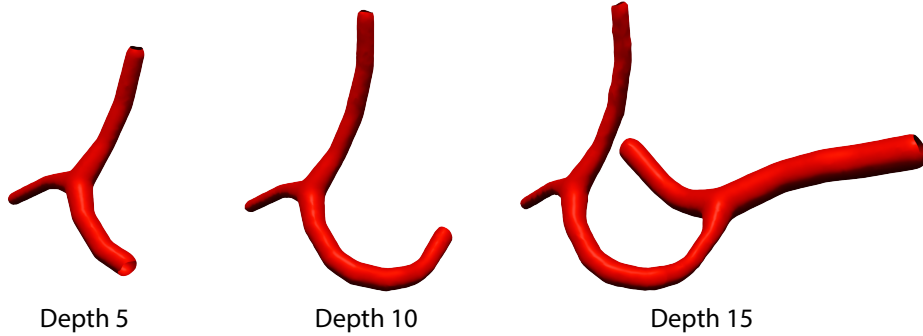


Figure 4.1: Representation of tree structures at varying heights: The same vascular tree trimmed to different heights (5, 10, and 15), demonstrating the structural changes and complexity reduction at lower heights.

4.1.2 Classifier training strategy

To calculate the L_{recon} loss term, we bypassed the classifier’s predictions and used the original tree topology for decoding during training. This approach allows us to compare node attributes effectively, even if the classifier fails to make accurate topology predictions while being trained to predict the degree of each node. Given the imbalance in node distribution across classes, we adjust the weights in the cross-entropy loss calculation by using the inverse of each class count. During preliminary experiments, we observed that accurately classifying nodes closer to the tree root is critical. A misclassification at the top levels (closer to the root) can lead to a cascading effect, impacting the reconstruction of all subsequent branches in the tree (i.e. skip reconstructing a branch). To account for this, we implemented and evaluated different weighting schemes assigning a weight to the cross-entropy loss based on the node depth and number of total sub-tree nodes.

4.1.3 Implementation details

We developed the data preprocessing pipeline and network code using Python, VMTK and the PyTorch Framework. For centerline extraction, we set the VMTK script’s advancement ratio to 1.05. This produces multiple cross-sections at bifurcations, where we select the one with the smallest radius to ensure alignment with the principal centerline

direction. All tree attributes were normalized to the range $[0, 1]$. In our training setup, the batch size was set to 4, and we used the ADAM optimizer with settings $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a learning rate of 1×10^{-4} . For Eq. 3.4, we used $\alpha = 0.3$ and $\gamma = 0.001$. All models were trained for 20,000 epochs.

To accelerate computation, we implemented dynamic batching [53], which groups operations for input trees of varying shapes and different nodes within a single graph. Training time is predominantly influenced by the height of the tree, as deeper trees require more sequential processing steps. Due to the implementation of dynamic batching, all nodes at the same depth are processed simultaneously, reducing the number of sequential steps required to transverse the tree. Consequently, while the amount of ramifications has some impact, it is relatively minor compared to the effect of tree height on training time. This is particularly noticeable with the Aneurisk dataset due to its larger number of ramifications compared to the Intra dataset, see Tab. 4.1 for more details about the amount of bifurcations and nodes for each dataset. On average, training our models takes between 12 and 36 hours. All experiments were run on a workstation equipped with an NVIDIA A100 GPU, 80GB VRAM, and 256GB RAM.

4.1.4 Experiments

We trained our generative autoencoders using two different datasets, two different resampling rates, and three maximum heights to investigate how these factors and different hyperparameter choices influence the model’s behavior and generative capabilities. After preprocessing and discarding vessels with loops, each dataset consisted of 100 samples. We applied resampling to the original trees to better capture their structure with a reduced number of nodes. This was achieved using the Ramer-Douglas-Peucker algorithm [21], [77]. Although resampling can affect the height of the original tree, we imposed a fixed maximum during training (5, 10, or 15) to ensure consistency across experiments. The resampling rate, defined by parameters $\epsilon = 0.1$ and $\epsilon = 0.2$, controls the maximum allowable distance between the original and the simplified curve, influencing how well the tree’s geometry is captured within the specified node limit. Additionally, we conducted an ablation study to assess the impact of the weighting scheme on the node classifier behavior.

Height	ϵ	Nodes			Bifurcations
		Min	Max	Avg	Avg
Aneurisk ([3])					
5	0.2	5	18	10.01	1.86
	0.1	5	18	9.65	1.53
10	0.2	12	43	25.00	3.38
	0.1	10	46	23.02	2.62
15	0.2	22	73	41.56	4.80
	0.1	15	73	38.44	3.73
Intra ([121])					
5	0.2	6	15	9.67	1.52
	0.1	7	15	9.78	1.46
10	0.2	6	30	14.53	1.69
	0.1	7	35	18.86	1.66
15	0.2	6	34	14.88	1.70
	0.1	7	46	20.64	1.72

Table 4.1: Node and bifurcation statistics for the Aneurisk and Intra datasets, computed at different tree heights and resampling rates (ϵ).

4.2 Results

We now evaluate our method against recent approaches (Sec. 4.2.1), beginning with a comparison with the methods of [90] and [42] in Sec. 4.2.1. This is followed by an analysis of results relative to the older baselines proposed by [31] in Sec. 4.2.1. In Sec. 4.2.2, we further explore the synthesis of pathological geometry features within vascular structures. Lastly, Sec. 4.2.3 presents an ablation study to evaluate the impact of relevant training configuration choices.

4.2.1 Comparison against baseline methods.

Sinha et al. [90] and Kuipers et al. [42]

We quantitatively evaluated the generated samples and compared them to recent methods using three established metrics: Minimum Matching Distance (MMD), Covergae (Cov), and 1-Nearest Neighbor Accuracy (1-NNA).

MMD [1] measures the quality of the generated samples by computing, for each real sample, the distance to its closest counterpart in the generated set and averaging these

	MMD ↓	COV ↑	1-NNA
Aneurisk			
Sinha et al.[90]	1.264	.19	.03
Kuipers et al.[42]	.537	.03	.0
VesselVAE (Ours)	.004	.58	.68
Intra			
Sinha et al.[90]	.447	.22	.1
Kuipers et al.[42]	.288	.04	.0
VesselVAE (Ours)	.039	.47	.4

Table 4.2: Quantitative results for novel tree generation comparing our method against baseline approaches. Metrics are denoted as follows: ↑ indicates that higher values are better, ↓ indicates that lower values are better, and for 1-NNA, the optimal score is 0.5. For [90] and [42], we observed low 1-NNA alongside low COV, indicating model collapse: the model produces highly similar samples that cluster in a narrow region of the distribution, sufficiently realistic to fool the 1-NN classifier, but lacking diversity.

values. Low MMD values indicate that the generated samples are close to the real ones, suggesting high fidelity and good coverage of the data distribution. COV [1] quantifies the diversity of the generated distribution by determining the proportion of real samples that can be paired with at least one generated sample. This is done by identifying the closest match in the reference set for each generated sample using a distance function such as Chamfer Distance (CD) or Earth Mover’s Distance (EMD). Higher COV values are desirable, as they indicate that a larger portion of the real distribution is represented by the generated set. Although coverage is effective in detecting mode collapse (e.g. when many generated samples map to the same or a few real samples), it does not provide information about sample quality. In some cases, perfect coverage can be achieved even if the matched pairs are significantly different. Finally, 1-NNA [120] evaluates the plausibility of the generated distribution by computing the leave-one-out 1-nearest neighbor classification accuracy between real and generated samples. A value close to 50% indicates that the two distributions are well-mixed, while scores significantly above or below this threshold suggest mode collapse or poor generalization.

Our evaluation was performed on both datasets, resampled at a rate of 0.2 and with a depth of 10, as this configuration yielded the best performance in our ablation study. The

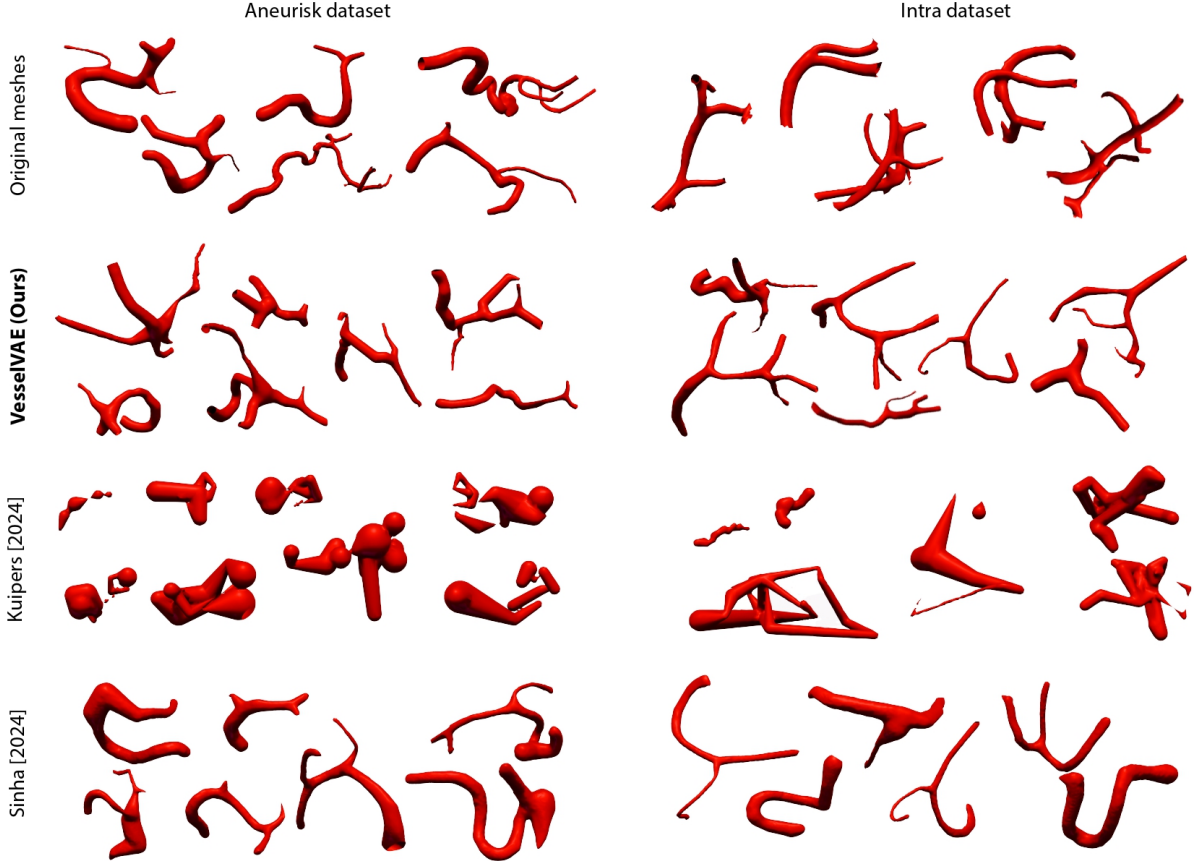


Figure 4.2: Comparison of vessel geometries generated by VesselVAE (Ours) and baseline methods trained on the Aneurisk and Intra datasets. The approach by [42] struggles with intricate geometry, often producing disconnected segments and failing to capture the underlying feature distribution. Meanwhile, [90] generates proper vessel geometries but lacks diversity, resulting in simpler and shallower structures. In contrast, our method demonstrates superior performance, producing diverse and realistic vessel meshes.

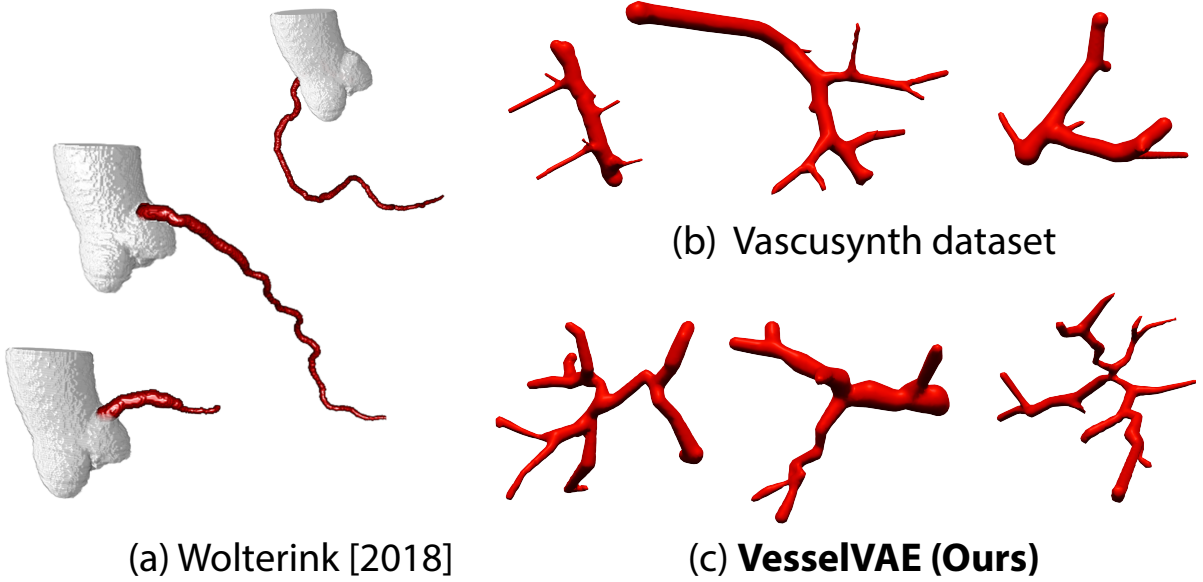


Figure 4.3: (a) Vessel geometries generated by [118]. The evaluation is limited to a visual comparison due to the lack of access to the original code or data. (b) Vessel geometries generated using the method presented in [31]. (c) Vessel geometries generated by our method when trained on (b). While occasionally struggling with unrealistic angles, VesselVAE effectively captures the underlying characteristics of the data, demonstrating a similar expressive power to the parametric method proposed by [31].

results are presented in Tab. 4.2. Our model consistently outperforms both baselines across all three metrics when trained on the same datasets. Additionally, Fig. 4.2 complements the quantitative analysis with example renders for all methods and training datasets. In terms of efficiency, our method is significantly more lightweight, using ~ 1 million parameters compared to the ~ 43 million and ~ 41 million parameters required by [90] and [42], respectively. Furthermore, our model has a much smaller memory footprint during training, requiring ~ 67 MB compared to ~ 173 MB and $\sim 2,000$ MB for the other methods.

Hammarneh et al. [31]

Comparing synthetic samples with real datasets (such as Aneurisk or Intra) is possible by measuring differences in parameter distributions. However, it is not trivial to compare our approach with other methods such as Vascusynth. This is because procedural and

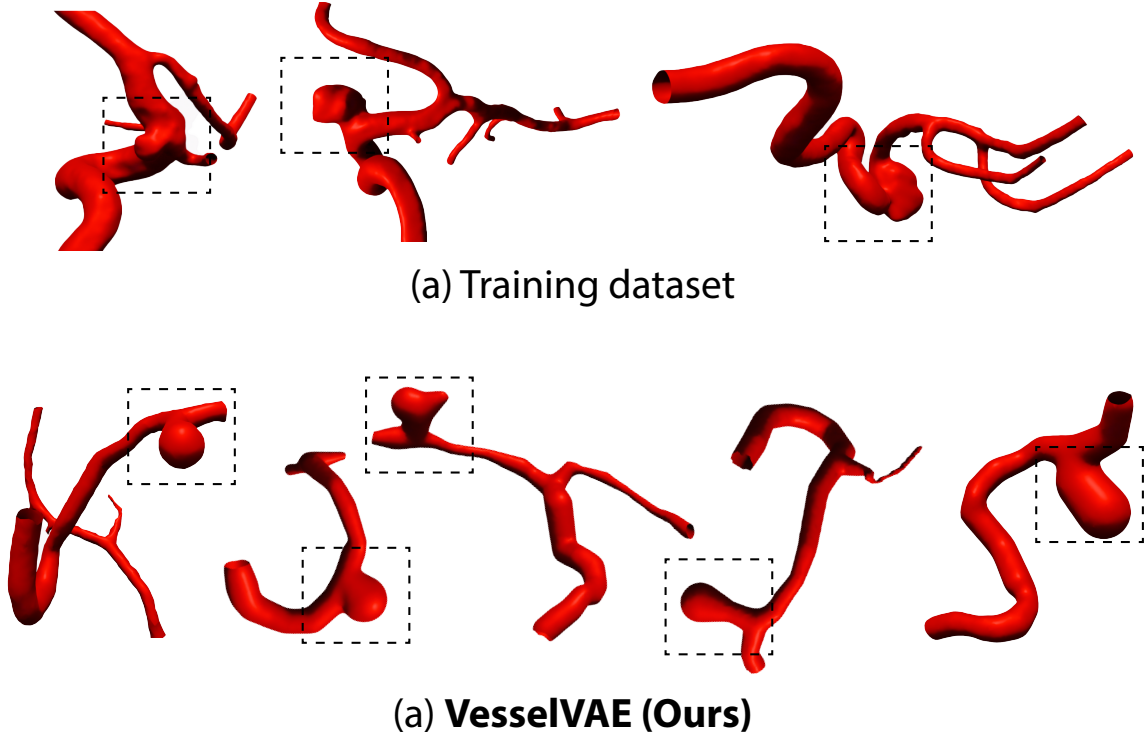


Figure 4.4: (a) Vessels with aneurysms from the Aneurisk dataset. (b) Vessels with aneurysms generated by VesselVAE.

data-driven methods have fundamentally different approaches, making direct comparison challenging.

To make this feasible, we generated a synthetic dataset using the method described in [31] and trained VesselVAE on these data. The goal is to assess whether VesselVAE can generate similar samples while maintaining the expressiveness and precision of the original generative method. In our experiments, the training dataset was resampled with $\epsilon = 0.1$ and $\epsilon = 0.2$ (as with previous comparisons). After training, we computed radius, tortuosity, and length metrics to compare distributions, see Tab. 4.4. Fig. 4.3 illustrates vessel geometries generated by VesselVAE when trained on Vascusynth data, demonstrating its ability to capture the main geometric characteristics of Vascusynth.

4.2.2 Pathology geometry synthesis

A key strength of our data-driven approach is its ability to naturally capture and reproduce distinctive geometric features present in the training data. In Fig. 4.4 we show some

synthesized vessel geometries featuring aneurysms after training on the Aneurisk dataset. Unlike parametric methods which require explicit coding to replicate such features, our approach inherently learns and reproduces both realistic non-pathological and pathological vessel structures.

4.2.3 Ablation

We conducted an ablation study to evaluate the impact of various configurations on our model’s performance. Specifically, we tested three different loss weighting schemes: (1) assigning a *uniform* weight of one to all nodes in the tree, (2) weighting each node based on its *depth* within the tree, and (3) weighting each node according to the size of its *sub-tree*, where each node is treated as the root of its respective sub-tree. Additionally, we explored the effects of two different vessel centerline resampling rates $\epsilon = 0.1$ and $\epsilon = 0.2$ and retrained our models using three different maximum heights (5, 10, and 15). This comprehensive evaluation allowed us to analyze the contributions of each factor to the overall performance of the model and select the best-performing configuration for the aforementioned comparisons.

To assess the best-performing model we established a set of metrics: tortuosity per branch, total vessel centerline length, and average tree radius. These metrics have been widely used in the field of 3D modeling of blood vessels, and have been shown to provide reliable and accurate quantification of blood vessels’ main characteristics [13], [44], [90]. The tortuosity distance metric [14], a measure of branch twistiness computed as the ratio between each branch’s length and the straight-line distance between its endpoints, has clinical importance in blood vessel analysis. We also evaluated total vessel length and average radius metrics previously used to differentiate healthy from cancerous malformations [13], [44]. Due to the inherent randomness in the generation process, we synthesized 10 distinct sets of 100 samples each for every combination of weights, dataset, depth, and resampling rate. Metrics were computed for each training and generated set, and we compared distributions using Cosine Similarity (CS) and Earth Mover’s Distance (EMD). We report the mean values along with 95 % confidence intervals in Tab. 4.3.

We observed high similarity scores between histograms at depths 5 and 10, indicating that the generated blood vessels are realistic. The slight decrease in similarity at depth 15, particularly for the Aneurisk dataset, is attributed to increased computational demands (see Sec. 4.3). The uniform scheme significantly deteriorates the similarity values across all

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Weighting Scheme	Max. Height	ϵ	Cosine Similarity $\uparrow$						Earth Mover Distance $\downarrow$					
			Radius		Tortuosity		Length		Radius		Tortuosity		Length	
			M	CI	M	CI	M	CI	M	CI	M	CI	M	CI
Aneurisk ([3])														
Uniform	5	0.2	.93	(.91 .95)	.88	(.86 .89)	.95	(.93 .97)	.55	(.47 .54)	2.53	(1.99 3.06)	.65	(.48 .82)
Depth			.91	(.93 .96)	.91	(.90 .94)	.90	(.95 .97)	.39	(.28 .49)	1.81	(1.44 2.29)	.45	(.28 .61)
Sub-tree			.91	(.92 .96)	.91	(.91 .94)	.90	(.93 .98)	.43	(.33 .52)	2.21	(1.81 2.66)	.28	(.24 .32)
Uniform		0.1	.94	(.93 .96)	.77	(.72 .83)	.93	(.91 .95)	.49	(.35 .62)	3.1	(2.42 3.79)	.64	(.47 .8)
Depth			.91	(.90 .94)	.81	(.79 .86)	.90	(.94 .97)	.54	(.4 .68)	2.41	(2.13 2.86)	.43	(.33 .53)
Sub-tree			.91	(.91 .94)	.9	(.87 .92)	.91	(.95 .96)	.54	(.41 .66)	2.61	(1.89 3.34)	.39	(.28 .5)
Uniform	10	0.2	.95	(.93 .96)	.86	(.83 .89)	.92	(.9 .94)	.68	(.54 .82)	2.37	(1.99 2.75)	.82	(.67 .97)
Depth			.91	(.90 .95)	.91	(.91 .96)	.90	(.94 .97)	.58	(.44 .72)	1.21	(1.08 1.34)	.38	(.22 .54)
Sub-tree			.91	(.94 .98)	.91	(.92 .95)	.91	(.95 .97)	.58	(.47 .69)	1.11	(.85 1.45)	.36	(.28 .44)
Uniform		0.1	.91	(.9 .93)	.77	(.7 .83)	.86	(.82 .9)	.74	(.55 .94)	3.08	(2.59 3.56)	1.27	(1.12 1.42)
Depth			.91	(.93 .95)	.91	(.88 .95)	.90	(.95 .97)	.49	(.42 .56)	1.5	(1.22 1.79)	.59	(.46 .71)
Sub-tree			.90	(.95 .97)	.91	(.9 .95)	.91	(.93 .97)	.47	(.36 .58)	1.6	(1.15 2.05)	.46	(.29 .63)
Uniform	15	0.2	.83	(.8 .86)	.83	(.81 .85)	.91	(.88 .93)	1.25	(1.14 1.31)	1.92	(1.37 2.48)	1.4	(1.18 1.62)
Depth			.91	(.9 .94)	.81	(.78 .87)	.40	(.4 .51)	.6	(.47 .72)	2.61	(1.87 3.4)	2.71	(2.54 2.91)
Sub-tree			.91	(.9 .95)	.91	(.94 .96)	.91	(.94 .95)	.54	(.42 .65)	1.11	(.96 1.41)	.81	(.71 .91)
Uniform		0.1	.94	(.92 .96)	.74	(.68 .8)	.86	(.84 .88)	.76	(.64 .88)	3.52	(2.82 4.22)	1.6	(1.44 1.76)
Depth			.91	(.92 .95)	.81	(.8 .88)	.71	(.69 .76)	.63	(.52 .73)	2.11	(1.55 2.71)	1.9	(1.8 2.02)
Sub-tree			.91	(.92 .95)	.81	(.81 .86)	.91	(.88 .94)	.77	(.65 .88)	1.71	(1.32 2.26)	.75	(.64 .85)
Intra ([121])														
Uniform	5	0.2	.97	(.96 .98)	.93	(.92 .94)	.93	(.9 .95)	.55	(.42 .67)	2.38	(1.79 2.97)	.48	(.4 .55)
Depth			.91	(.94 .96)	.91	(.91 .95)	.91	(.91 .95)	.56	(.46 .65)	1.7	(1.3 2.09)	.39	(.32 .46)
Sub-tree			.91	(.92 .96)	.91	(.91 .96)	.91	(.92 .96)	.42	(.29 .56)	1.81	(1.41 2.32)	.52	(.42 .63)
Uniform		0.1	.97	(.96 .98)	.91	(.89 .95)	.95	(.93 .96)	.44	(.35 .53)	2.7	(2.08 3.32)	.42	(.35 .5)
Depth			.9	(.88 .91)	.9	(.87 .94)	.91	(.95 .97)	.59	(.53 .65)	2.61	(1.96 3.31)	.52	(.4 .64)
Sub-tree			.91	(.96 .98)	.9	(.87 .93)	.91	(.93 .97)	.51	(.42 .6)	3.41	(2.3 4.69)	.39	(.33 .45)
Uniform	10	0.2	.95	(.94 .97)	.89	(.87 .92)	.93	(.9 .96)	.66	(.52 .79)	2.33	(1.66 3.0)	.41	(.33 .49)
Depth			.91	(.95 .99)	.91	(.95 .98)	.91	(.94 .97)	.41	(.37 .45)	1.51	(1.06 1.98)	.42	(.32 .53)
Sub-tree			.91	(.93 .97)	.91	(.9 .95)	.90	(.93 .98)	.52	(.42 .61)	1.6	(1.32 1.88)	.31	(.23 .39)
Uniform		0.1	.96	(.94 .97)	.81	(.77 .85)	.95	(.94 .97)	.39	(.34 .43)	2.81	(1.85 3.78)	.68	(.61 .76)
Depth			.91	(.93 .96)	.91	(.93 .97)	.91	(.92 .96)	.56	(.4 .73)	1.31	(1.0 1.78)	.68	(.55 .8)
Sub-tree			.91	(.92 .96)	.80	(.81 .9)	.91	(.92 .97)	.54	(.43 .66)	2.01	(1.74 2.34)	.41	(.25 .56)
Uniform	15	0.2	.94	(.92 .96)	.9	(.88 .93)	.92	(.89 .94)	.67	(.51 .83)	2.11	(1.59 2.62)	.54	(.39 .69)
Depth			.91	(.93 .97)	.90	(.94 .97)	.91	(.9 .94)	.43	(.36 .49)	1.01	(.84 1.31)	.48	(.37 .59)
Sub-tree			.90	(.95 .97)	.91	(.91 .96)	.90	(.95 .98)	.38	(.29 .47)	1.61	(1.33 1.93)	.38	(.29 .47)
Uniform		0.1	.95	(.94 .97)	.88	(.83 .93)	.95	(.93 .97)	.65	(.53 .78)	2.71	(1.85 3.57)	.93	(.75 1.11)
Depth			.90	(.93 .98)	.91	(.9 .96)	.91	(.94 .97)	.54	(.44 .65)	2.6	(2.14 3.06)	1.0	(.82 1.19)
Sub-tree			.91	(.92 .96)	.80	(.81 .91)	.91	(.93 .96)	.67	(.56 .79)	1.61	(1.14 2.14)	.56	(.42 .69)

Table 4.3: Ablation summary. Cosine Similarity and Earth Mover Distance for all the evaluated metrics with resampling parameter ϵ set to 0.1 and 0.2. All experiments were carried out 10 times with randomized seeds. We report the mean values (M) and 95 % confidence intervals (CI). The weighting scheme column corresponds to loss weight ablation. *Uniform* refers to constant weighting, *Depth* indicates depth-based weighting, and *Sub-tree* corresponds to sub-tree size weighting. In bold the best-performing scheme for trees of depth 10 and resampling rate 0.2. These highlighted models were used for the evaluation in Tab. 4.2 and Fig. 4.2

evaluated metrics. The effect of non uniform weighting gains importance when the height of the tree increases, while for shallow trees (height 5 in our experiments) performance remains mostly unaffected. Fig. 4.5 shows the distribution for the evaluated metrics of the best-performing models for trees of maximum height 10.

To identify the best-performing scheme, we computed the average cosine similarity across all metrics—Radius, Tortuosity, and Length—for each row in Table 4.3. We then averaged the cosine similarity for each weighting scheme, resulting in a single similarity value per scheme in Tab. 4.4

Dataset	Uniform	Depth	Sub-tree
Intra	0.928	0.938	0.933
Aneurisk	0.87	0.87	0.93

Table 4.4: Average cosine similarity across all metrics (Radius, Tortuosity, and Length) for each weighting scheme.

In the Aneurisk dataset, the sub-tree size weighting scheme exhibited superior performance, while in the Intra dataset, the depth-based weighting scheme achieved slightly better results compared to the others. This variation is likely attributable to the greater number of ramifications in the Aneurisk dataset (see Tab. 4.1 for dataset statistics), which has a stronger influence on the computation of loss weights. Conversely, the Intra dataset, with its fewer ramifications, showed comparable performance across both schemes. This highlights the role of this weight parameter in the autoencoder’s capability to accurately synthesize vasculature geometry.

4.3 Discussion and Limitations

Our experiments utilized two open-source datasets of cerebral vasculature, resampled at different rates to show the impact on evaluation metrics. Results indicate that complexity increases with tree height, leading to good results with lower tree depths and suboptimal outcomes at the highest depths tested. While we experimentally found an optimal bottleneck size in our experiments, we hypothesize that the recursive encoding framework may lose information with high hierarchies. A per-branch evaluation could offer deeper insights (as proposed by Kuipers et al.). However, it requires labeled data which is not available in our setting. Additionally, the generated geometries might show self-intersections. This issue arises because, although the model effectively generates plausible

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Vascusynth														
Weighting Scheme	Max. Height	ϵ	Cosine similarity $\uparrow$						Earth Mover Distance $\downarrow$					
			Radius			Tortuosity			Radius			Tortuosity		
			M	CI	M	CI	M	CI	M	CI	M	CI	M	CI
Depth	5	0.2	.96	(.95 .97)	.51	(.47 .56)	.80	(.74 .85)	.42	(.32 .53)	2.67	(1.83 3.52)	.44	.3 .59
Sub-tree			.9	(.89 .95)	.7	(.74 .83)	.9	(.91 .95)	.54	(.45 .64)	3.3	(2.68 4.01)	.46	(.33 .58)
Depth		0.1	.92	(.89 .94)	.87	(.83 .91)	.74	(.64 .83)	.41	(.3 .52)	3.64	(2.85 4.43)	1.05	(.91 1.19)
Sub-tree			.5	(.88 .92)	.6	(.55 .64)	.8	(.82 .87)	.65	(.52 .85)	5.4	(4.33 6.47)	1.7	(1.59 1.94)
Depth	10	0.2	.87	(.85 .89)	.93	(.91 .96)	.88	(.85 .9)	.61	(.42 .8)	1.95	(1.43 2.47)	.46	(.37 .55)
Sub-tree			.9	(.88 .92)	.8	(.83 .86)	.9	(.95 .97)	.97	(.7 1.23)	1.1	(.93 1.44)	.85	(.71 .98)
Depth		0.1	.96	(.95 .97)	.91	(.89 .93)	.83	(.81 .85)	.5	(.38 .62)	2.68	(1.86 3.49)	.66	(.54 .79)
Sub-tree			.8	(.88 .9)	.6	(.55 .71)	.8	(.79 .84)	1.0	(0.9 1.11)	3.3	(1.91 4.76)	1.5	(1.42 1.72)
Depth	15	0.2	.5	(.47 .53)	.65	(.59 .72)	.56	(.49 .63)	2.65	(2.34 2.96)	1.38	(1.02 1.74)	2.9	(2.53 3.26)
Sub-tree			.8	(.79 .85)	.7	(.68 .84)	.8	(.85 .89)	1.4	(1.28 1.53)	2.4	(1.51 3.3)	.77	(.64 .89)
Depth		0.1	.86	(.82 .9)	.82	(.79 .85)	.74	(.68 .79)	.87	(.79 .94)	1.41	(1.02 1.79)	3.69	(3.07 4.32)
Sub-tree			.7	(.71 .78)	.4	(.26 .6)	.4	(.38 .47)	1.9	(1.82 2.14)	2.8	(1.68 3.94)	2.9	(2.8 3.13)

Table 4.4: Cosine Similarity and Earth Mover Distance for all the evaluated metrics with resampling parameter $\epsilon=0.1$ and $\epsilon=0.2$, with trees trimmed to depths 5, 10 and 15. We run the experiments 10 times with random initial vectors and reported the mean values (M) and 95 % confidence intervals (CI) for VesselVAE trained on synthetic data generated with [31]. We attribute the larger performance drop on Vascusynth, especially at greater depths to a mismatch between VesselVAE’s inductive biases and the synthetic nature of the data. The VesselVAE architecture was designed and tuned for anatomical realism based on real datasets (Aneurisk and Intra), whereas Vascusynth is generated via a parametric algorithm. We hypothesize, the model is being pushed to generalize beyond the structural patterns it was intended to capture.

geometries, it does not enforce strict constraints to eliminate self-intersections. In the future, we would like to incorporate restrictions into the generative model to avoid such artifacts. Furthermore, while our method synthesizes vessels highly similar to those in the training dataset, some exhibit implausible angles. This suggests the model struggles with angle learning. Potential improvements include adding an attribute for angles, exploring different angle representations, or resampling arteries with greater detail on curves. Examples of failed cases can be seen in Fig. 4.6.

Our method enables efficient encoding and decoding of binary tree structures and produces high-quality synthesized models. Our findings highlight the versatility of our network architecture, demonstrating its ability to synthesize blood vessels across different datasets, including those with aneurysms. Additionally, our approach outperforms recent state-of-the-art methods by more accurately replicating intricate vascular characteristics while producing more realistic and diverse geometries. Overall, our proposed approach

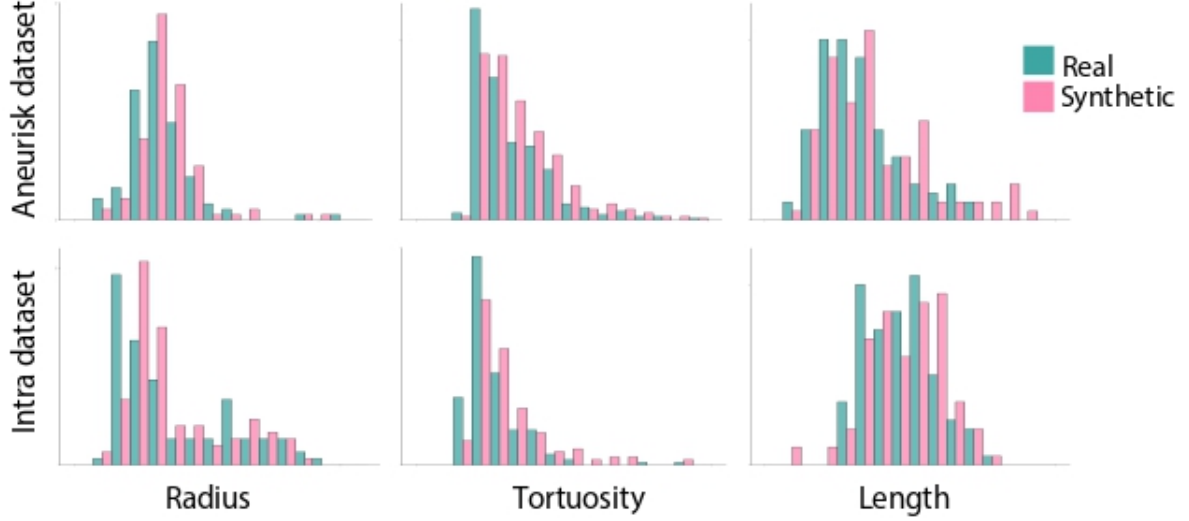


Figure 4.5: Quantitative evaluation of generated vessel samples. The histograms display the distributions of each evaluated metric. Data generated by VesselVAE is labeled as *synthetic* (pink), while data from the datasets is labeled as *real* (blue). All histograms are computed from vessels with maximum height 10 and resampling $\epsilon \in 0.2$

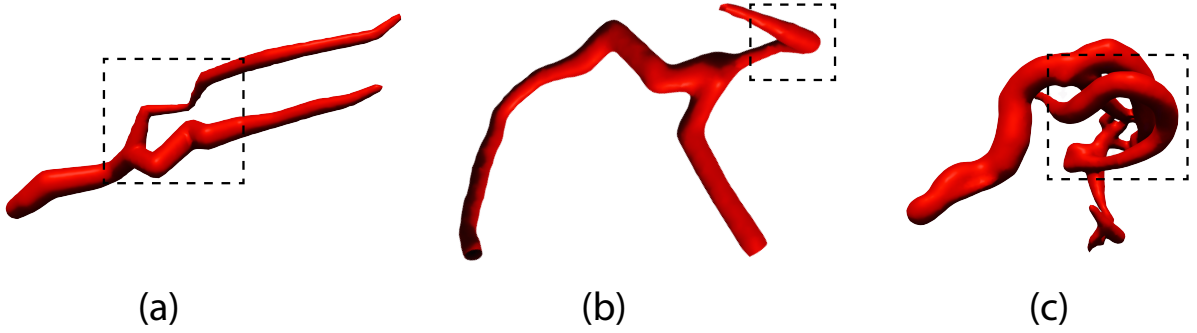


Figure 4.6: Examples of VesselVAE generations exhibiting implausible angles (a, b) and self-intersections (c).

shows potential for improving 3D blood vessel geometry synthesis and supporting the development of clinical tools that could benefit healthcare professionals.

Chapter 5

VesselGPT: Autoregressive Modeling of Vascular Geometry

The previous method, VesselVAE, introduced in Chapters 3 and 4, is a recursive variational neural network (RvNN) tailored for vascular geometry generation. However, like other parameterized methods, it represents vessels solely by their centerlines and radii. While this compact representation is widely used, it oversimplifies the vessel’s cross-sectional geometry, potentially leading to inaccuracies in simulations based on such data [24]. Our data exploration revealed that vessel cross-sections are far from being circular, confirming the need for a non circular parametrization. Moreover, the method is limited to relatively small vascular segments, a constraint that arises from the difficulty the RvNN architecture faces when scaling to deeper levels of recursion.

In light of these limitations, we propose a novel Transformer-based [112] approach for blood vessel synthesis. Although transformers have shown remarkable versatility in modeling sequential data, including geometric structures [88], [98], their application to vascular generation remains largely unexplored. In this work we present a formulation that treats vascular trees as a sequence of nodes, leveraging the GPT-2 architecture to generate compact, anatomically realistic vessel structures. Additionally, we propose using a cross-sectional representation of the vessels’ channels based on B-splines. This approach allows a more nuanced surface model compared to methods that rely solely on a single radius value, overcoming a key limitation in previous techniques. Our approach proceeds in two stages. First, we use a vector-quantized autoencoder (VQ-VAE) [108] to learn a discrete

vocabulary of node attributes, capturing essential geometric features in context. We then employ this learned codebook to train a GPT-2 model [74], enabling the autoregressive generation of 3D blood vessel structures. Finally, we show experiments comparing our method with baseline techniques, validating the effectiveness of our approach.

5.1 Quantitative Analysis of Vessel Cross-Sectional Shape

During the development of our framework, exploratory data analysis revealed that vascular cross-sections deviate substantially from the idealized circular assumption commonly adopted in prior work. This simplification, while computationally convenient, can introduce systematic inaccuracies in downstream analyses such as computational fluid dynamics or morphological assessment. The impact of this idealization has been explicitly investigated in prior studies [24], which demonstrated that enforcing circular shapes can significantly alter hemodynamic predictions.

To quantitatively assess this deviation, we compared actual vessel cross sections from the Aneurisk dataset with several geometric approximations:

- (1) our proposed *B-spline representation*, obtained by fitting a parametric B-spline curve to the ordered contour points to capture smooth, irregular boundaries;
- (2) an *average-fitting circle*, obtained by computing the cross-sectional area and deriving the equivalent radius under the assumption of a perfect circle. This is the parametrization used in our previous approach;
- (3) the *minimum enclosing circle*, defined as the smallest circle that fully contains the contour, minimizing the maximum radial distance; and
- (4) the *maximum inscribed circle*, defined as the largest circle entirely contained within the contour, maximizing the minimum radial distance from the centroid to the boundary.

The shape discrepancy between two contours A and B was evaluated using the bidirectional Hausdorff distance. This distance is defined as *the maximum distance of a set to the nearest point in the other set* [84]. More formally, the *directed Hausdorff distance* from

set A to set B is given by:

$$h(A, B) = \max_{a \in A} \min_{b \in B} d(a, b),$$

where a and b are points of sets A and B , respectively, and $d(a, b)$ is a distance metric. In our work, we take $d(a, b)$ to be the Euclidean distance between a and b .

Table 5.1: Hausdorff distance (mean $\pm$ standard deviation) between actual cross-sections from the Aneurisk dataset and different geometric approximations. Lower values indicate better shape agreement.

Method	Mean	Std. Dev.
Spline representation	0.86	4.18
Average-fitting circle	15.11	16.88
Minimum enclosing circle	6.02	7.12
Maximum inscribed circle	11.38	12.90

Table 5.1 summarizes the mean and standard deviation of the Hausdorff distance across all cross-sections in the dataset. Our spline-based method consistently achieves the lowest average error, confirming its superior capability to capture complex and irregular cross-sectional shapes in vascular structures.

5.2 Methods

We propose a sequence-based approach for autoregressively generating blood vessel trees as sequences of nodes (Figure 5.1). The proposed method consists of two main stages:

1. A Vector-Quantized Variational Autoencoder (VQ-VAE) trained to learn a discrete vocabulary of tokenized node embeddings.
2. A transformer model trained for vessel tree generation by performing autoregressive next-node prediction over the learned vocabulary of tokens.

5.2.1 Codebook learning

Autoregressive generative models produce token sequences by conditioning the generation of each new token on those previously produced, thereby capturing complex dependencies within the sequence. To apply this paradigm to blood vessel synthesis, we first convert

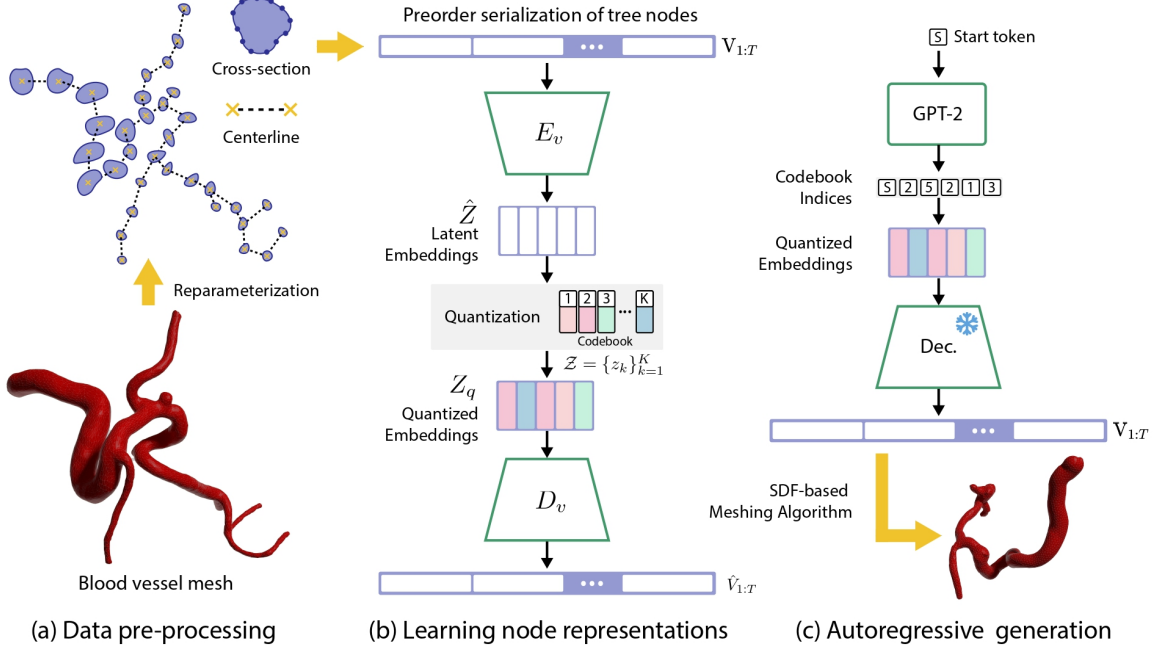


Figure 5.1: VesselGPT overview. (a) We begin by reparametrizing the blood vessel meshes by computing their centerlines and fitting B-splines to the cross-sections. We store all parameters in a binary tree, where each node represents a centerline sample along with the corresponding weights of its cross-section spline. (b) We flatten the tree using pre-order traversal, then feed the resulting sequence into a Vector Quantized Variational Autoencoder (VQ-VAE). The encoder maps it to a discrete latent space, and the decoder learns to reconstruct the original sequence from the quantized embeddings. (c) We train a GPT model on the codebook index sequences (tokens) that the pretrained VQ-VAE encoder generates for each dataset sample. During inference, the autoregressive model generates token sequences, which are decoded back to the serialized tree data structured by means of the VQ-VAE decoder. Finally, a meshing algorithm reconstructs the vessel geometry from its parameterized representation.

each vascular structure into a one-dimensional sequential representation that can be directly processed by the model.

We represent each vessel as a binary tree, where every node corresponds to a vessel segment, and its attributes encode geometric information: 3D coordinates of the centerline point and cross sectional spline coefficients. To linearize this hierarchical structure, we perform a *preorder traversal*, a tree-walking strategy that visits each node before its children (*root* $\rightarrow$ *left child* $\rightarrow$ *right child*). This choice preserves the hierarchical context

of each node in the resulting sequence, allowing the autoregressive model to naturally condition the generation of a child vessel segment on its parent’s attributes.

The preorder traversal produces an ordered vector of node attributes, which we treat as the raw sequence. For absent child nodes that occur when a node has less than two children, we insert a special *zero attribute vector*, ensuring that the complete original tree topology can be reconstructed during decoding.

Using the coordinates and spline coefficients directly as tokens for autoregressive generation is not optimal, mainly because it fails to effectively capture geometric patterns. This is because such a representation lacks information about neighboring vessel nodes and does not incorporate any priors from the distribution of vascular structures. To address this challenge, we propose to learn a quantized embedding from a collection of vessel trees. The task formulation is specified as follows: Let $V_{1:T} = (v_1, \dots, v_T)$ represent a sequence of ground-truth blood vessel nodes, where each node $v_t \in \mathbb{R}^m$ is associated with an m -dimensional attribute vector that includes its spatial coordinates x, y, z along with the coefficients and control points defining the spline. The goal of this task is to sequentially predict the synthesized blood vessel nodes $\hat{V}_{1:T}$, approximating the original sequence $V_{1:T}$.

To construct the node vocabulary, we employ a vector-quantized autoencoder (VQ-VAE)[108], which consists of an encoder (E_v), a decoder (D_v), and a discrete codebook $\mathcal{Z} = \{z_k\}_{k=1}^K$, where $z_k \in \mathbb{R}^{d_z}$, and d_z is the dimension of the codebook vectors. The VQ-VAE is trained to self-reconstruct realistic vessel trees. Specifically, given vessel tree data in the continuous domain $V_{1:T}$, the VQ-VAE encoder (E_v), which incorporates a transformer layer [112], first encodes the continuous sequence of vessel nodes into latent features $\hat{Z} \in \mathbb{R}^{T \times d_z}$. These latent features $\hat{Z}$ are then quantized to Z_q via an element-wise quantization function Q_v , which maps each feature vector to its closest entry in the codebook:

$$Z_q = Q_v(\hat{Z}) := \arg \min_{z_k \in \mathcal{Z}} \|\hat{z}_t - z_k\|_2 \quad \forall t \in [1, T]. \quad (5.1)$$

The quantized representation Z_q is subsequently reconstructed into continuous vessel tree representations $\hat{V}_{1:T}$ by the VQ-VAE decoder (D_v), which has a symmetric architecture

to the encoder. The VQ-VAE model is trained using the following objective function:

$$\mathcal{L}_{\text{VQ}} = \|V_{1:T} - \hat{V}_{1:T}\|_1 + \|\text{sg}(\hat{Z}) - Z_q\|_2^2 + \lambda \|\hat{Z} - \text{sg}(Z_q)\|_2^2, \quad (5.2)$$

where the first term enforces reconstruction consistency, the second and third terms update the codebook via the stop-gradient operator sg [8], and λ is a weighting factor.

5.2.2 Autoregressive vessel sequence prediction

The second component in our model is a GPT-style decoder-only transformer that takes advantage of the quantized embedding from the previous stage. Given a sequence of tokens computed from the nodes of a vessel tree, the transformer is trained to predict the codebook index of the next embedding in the sequence. Once trained, the transformer can be auto-regressively sampled to predict sequences of embeddings. These tokens can then be decoded into the tree geometry of novel vessel trees.

The input of the transformer consists of the codebook indices of the embeddings $e(Z_q)$, obtained by encoding the input sequence using the frozen encoder of stage 1. The list of codebook indices is prefixed and suffixed with predefined start and end tokens. The features then pass through a stack of multi-headed self-attention layers, where the transformer is trained to predict the next quantized codebook entry in the sequence. Specifically, we maximize the *log* probability of the training sequences with respect to the transformer parameters θ :

$$\prod_{t=1}^T p(e^t \mid e^{<t}; \theta), \quad (5.3)$$

where e^t represents the codebook index corresponding to the vessel node at step t , and $e^{<t}$ denotes the sequence of tokens up to step $t - 1$.

Once trained, the transformer can autoregressively generate a sequence of tokens; starting with a learned start token and continuing until a stop token is reached using beam sampling. The codebook embeddings indexed by the generated token sequence are then decoded by D_v to reconstruct the structure of the vessel.

5.2.3 3D mesh synthesis.

To reconstruct a 3D mesh representation of a blood vessel from a set of splines, we employ a three-step approach that involves centerline fitting, signed distance field (SDF) generation, and surface extraction through marching cubes. First, we fit a B-spline to each branch of the centerline and obtain a smooth parametric representation. Next, we construct a signed distance field (SDF) that encapsulates the vessel’s shape. The cross-sectional geometry of the vessel is characterized by a set of B-splines defining the radius at different points along the centerline. The SDF is then built by interpolating these cross-sectional shapes along the vessel’s trajectory, yielding a continuous representation of the vessel. Finally, we apply the marching cubes algorithm [54] to the generated SDF and obtain a triangulated surface mesh.

5.3 Experimental Setup

Materials. Our model was trained on the publicly available Aneurisk dataset¹ [3]. This dataset comprises 100 vessel segments reconstructed from 3D angiographic images, representing both healthy vasculature and aneurysms. We transformed the 3D meshes into binary tree representations, and obtained the vessel centerline with the *network extraction* script from the VMTK toolkit². The centerline points are obtained by calculating the ratio of the sphere step to the local maximum radius, using a user-defined advancement ratio. For each sample along the centerline, the contour of the blood vessel conduit was delineated by fitting a B-spline to the surface points located on the plane perpendicular to the centerline. Data augmentation was performed by resampling the trees at various rates and applying random rotations to the entire tree. Additionally, non-binary trees were converted to binary form, and any trees containing loops were excluded. This procedure ultimately yielded 528 binary trees from the original 3D meshes.

To serialize a binary tree, we traverse it in preorder and store each node’s attributes sequentially. The tree can be incomplete, meaning that some nodes have fewer than two children or none. To preserve the structure during deserialization, we insert zero vectors in place of these non-existing children, ensuring that the hierarchy can be accurately reconstructed. However, since the VQ-VAE learns an optimal representation for each

¹<https://github.com/permfl/AneuriskData>

²<http://www.vmtk.org/vmtkscripts/vmtknetworkextraction>

tree based on the entire dataset, these zero vectors are often mapped to different token sequences. As a result, after reconstructing the unquantized tree from the decoder, we threshold values lower than 10^{-2} to zero to ensure non-existing nodes are accurately predicted and the hierarchical tree structure can be reconstructed.

Non-binary trees were transformed into binary trees through an iterative algorithm designed to split multifurcations into binary bifurcations. For each node with more than two children, the algorithm introduces intermediate nodes with the same position and features as the original. These intermediate nodes are then used to sequentially divide the children into pairs, ensuring that no node has more than two children. The process carefully preserves the original spatial arrangement and attributes of the vessel, altering only the tree topology. We excluded vessel models with loops from our experiments.

Implementation details. For extracting the centerline, we configured the VMTK script with an advancement ratio of 1.05. The script generates several cross-sectional views at bifurcation points; in these instances, we opted for the cross-section with the smallest area to ensure it aligns correctly with the principal direction of the centerline. All attributes were normalized to the range $[-1, 1]$, and the root node of each tree was positioned at the origin.

Training. For Stage 1, we defined the codebook vector dimension as 64 and utilized 16 tokens per tree node. The batch size was set to 1, and we applied the ADAM optimizer with hyperparameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a learning rate of 1×10^{-4} . Training the model on a single NVIDIA RTX A4500 requires approximately 3 hours. For Stage 2, we maintained the same hyperparameters, learning rate, and optimizer. Training time for this stage is approximately 24 hours.

Metrics. We benchmarked our method with state-of-the-art approaches using established metrics for assessment, consistent with previous works [88], [90], [120]: We use Minimum Matching Distance (MMD) to evaluate quality, Coverage (COV) to measure diversity, and 1-Nearest Neighbor Accuracy (1-NNA) to assess plausibility. To compute these point-based metrics, we randomly sample 1000 points from all baseline generated meshes.

To further assess the quality of our generated samples, we compared their distribution to that of the training dataset using vessel-specific metrics. These metrics are well-established in 3D blood vessel modeling and have shown to effectively capture key vascular properties [13], [23], [44], [90]. Specifically, we evaluated branch-wise tortuosity

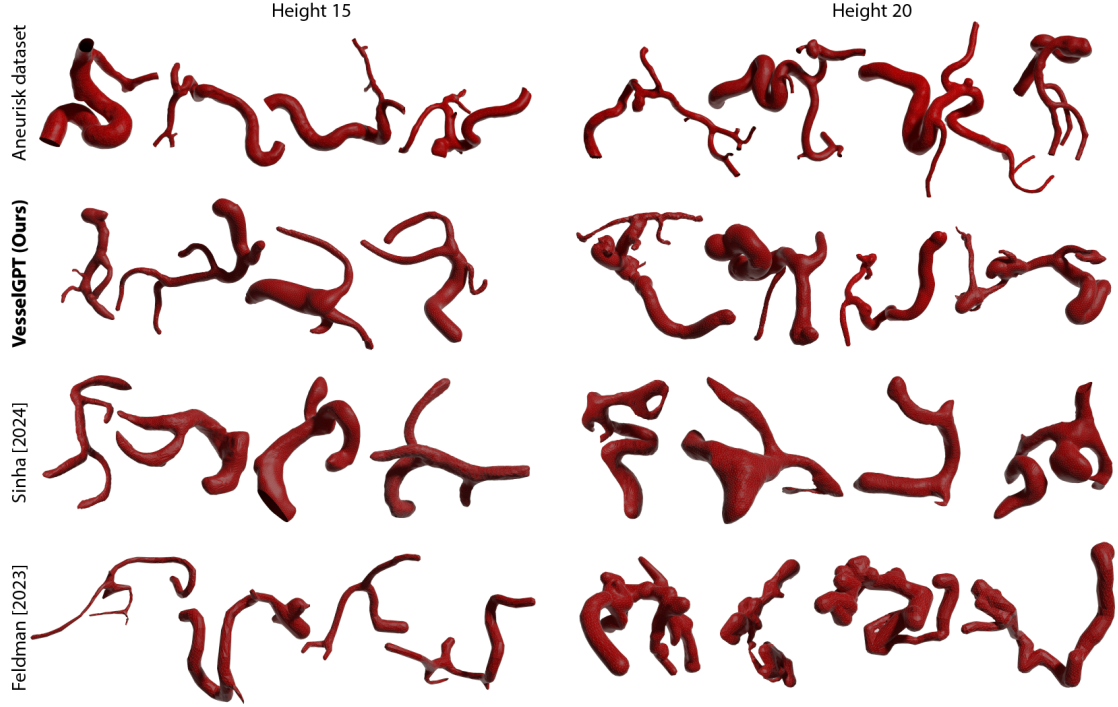


Figure 5.2: Comparison of vessel geometries generated by VesselGPT (Ours) and baseline methods trained on the Aneurisk dataset. Sinha et al. [90] produce valid vessels but lack diversity, resulting in simpler, shallower structures. Feldman et al. [23] performs well up to height 15 but struggles with deeper trees. In contrast, our method generates more diverse and realistic vessel meshes.

and total vessel centerline length. Tortuosity, measured using the distance metric [14], is particularly relevant due to its clinical significance, as it quantifies the twisting complexity of individual vessel branches. Total vessel length, on the other hand, has been widely used to distinguish between healthy and pathological vasculature. Finally, to assess how closely the generated vessel structures align with real ones, we quantify the similarity between their distributions using cosine similarity.

5.4 Results

5.4.1 Traditional 3D synthesis evaluation

We compare our method to leading vasculature generation approaches. Sinha et al. [90] uses diffusion over the space of INRs, focusing primarily on representation rather than the generation of novel structures. Feldman et al. [23] employs a recursive approach to generate vessel sequences but faces limitations with deeper trees and simplifies radius parametrization to a single value per centerline point, losing fine details. The results are presented in Tab. 5.2. Both our model and the baselines were trained with two different datasets: the Aneurisk trees trimmed at height 15 and 20. For the MMD metric, while a low value indicates resemblance, an extremely small value may suggest memorization. Our model outperforms the method by [90] in both COV and 1-NNA. Although VesselVAE achieves a higher COV score, a high COV value can be achieved with low-quality generated samples because the metric only measures the fraction of real samples that are the nearest neighbor to a generated sample [1]. This means that if the generated set broadly covers the range of real space, regardless of whether individual samples are anatomically plausible, the COV score can still be high. In other words, coverage reflects diversity but not fidelity: even unrealistic or noisy shapes can be close to some real sample, inflating the metric without guaranteeing meaningful anatomical realism. Qualitative evaluation further highlights VesselVAE’s difficulties with deeper trees and its tendency to oversimplify vessel radii using circular cross-sections, preventing it from capturing fine details. While these metrics are commonly used in the literature, they fail to capture key characteristics of the data. Furthermore, Fig. 5.2 provides example renders for all methods and training datasets, complementing the quantitative analysis.

5.4.2 Morphological metrics distributions comparison

Since traditional 3D synthesis evaluation metrics fail to capture the anatomical and physiological plausibility of generated vascular structures, we performed an additional assessment using domain-specific descriptors [13], [44]. In particular, we evaluated two widely used morphological features in vascular analysis: *total vessel length* and *tortuosity*.

For both the real and generated vessel sets, we computed the histograms of each descriptor and quantified their similarity using the *cosine similarity* metric. Given that all histogram

Table 5.2: Quantitative results for novel tree generation comparing our method against baseline approaches. $\uparrow$ indicates that higher values are better, $\downarrow$ indicates that lower values are better, and for 1-NNA, the optimal score is 0.5.

	Aneurisk Height 15			Aneurisk Height 20		
	MMD $\downarrow$	COV $\uparrow$	1-NNA	MMD $\downarrow$	COV $\uparrow$	1-NNA
Sinha et al.[90]	0.43	0.103	0.	0.418	0.104	0.
Feldman et al.[23]	0.014	0.49	0.238	.013	0.45	0.126
Ours	0.14	0.31	0.097	0.14	0.41	0.17

values are non-negative, cosine similarity values range from 0 (completely different) to 1 (identical distributions). Higher values indicate a closer match between the synthetic and real datasets.

We compared our approach against the previous VesselVAE [23] at two maximum tree heights ($h = 15$ and $h = 20$). As shown in Table 5.3, our method consistently achieved higher similarity scores for both tortuosity and length distributions.

Table 5.3: Cosine similarity between real and generated vessel descriptor distributions. Higher values indicate closer agreement. Best results for each height are in bold.

Model	h = 15		h = 20	
	Tortuosity	Length	Tortuosity	Length
VesselVAE [23]	0.46	0.83	0.77	0.38
Ours	0.96	0.87	0.97	0.88

These results indicate a strong agreement between the morphological statistics of real and generated vessels, demonstrating the realism of our synthesized geometries.

5.5 Conclusions

In this chapter we introduced VesselGPT, a novel method for generating blood vessels in an autoregressive manner by sequentially sampling from a learned codebook. Our method is capable of learning a discrete representation of blood vessels in an embedding space, that allows synthesizing high quality novel structures. We advance the state of the art by introducing a more compact yet expressive representation of vessel geometry, using B-splines to model the cross-sectional shape.

Chapter 6

Conclusiones y Trabajo Futuro

6.1 Conclusion

This thesis addressed the challenge of synthesizing realistic 3D vascular geometries by proposing two complementary generative frameworks: a recursive variational autoencoder (VesselVAE) and an autoregressive model based on tokenized discrete representations (VesselGPT). Both approaches advance the state of the art in vascular modeling by offering novel solutions for encoding structural complexity, generating diverse anatomical variants, and improving geometric fidelity.

VesselVAE introduced a recursive neural architecture capable of efficiently encoding and decoding vascular trees, demonstrating strong generalization across datasets, including those containing pathological cases such as aneurysms. In turn, VesselGPT offered an autoregressive approach to sequentially sample vessels, leveraging a compact representation via a codebook and incorporating anatomically meaningful priors, such as modeling cross-sections with B-splines.

Although VesselVAE outperforms VesselGPT on conventional quantitative metrics, this comparison also highlights the limitations of current evaluation strategies in vascular modeling. Standard metrics often fail to capture clinically relevant aspects such as anatomical plausibility or structural diversity. Nevertheless, VesselGPT exhibits several key advantages for real-world applications. In particular, it is significantly more efficient, requiring fewer training hours than VesselVAE. It also scales more effectively to larger and

deeper vascular trees—reaching heights of up to 20 compared to 15—and supports the modeling of more complex anatomical features, such as non-circular cross-sections, which are essential to accurately capture pathologies like aneurysms. These capabilities, although not fully reflected in standard metrics, underscore the need for more domain-specific evaluation methods aligned with the requirements of biomedical applications.

Together, these contributions illustrate the potential of combining structural priors with learned geometric representations, enabling the synthesis of anatomically coherent vessels. Importantly, both methods facilitate future integration into clinical workflows by enabling scalable generation of vascular models for data augmentation, simulation, and surgical planning. In summary, the models developed in this thesis provide a foundation for data-driven vascular synthesis and open the door to richer, more integrated tools for computational medicine and biomedical research.

The development of generative models for synthetic 3D vascular geometries, as demonstrated by VesselVAE and VesselGPT, is not merely an academic exercise; it addresses a critical bottleneck in computational biomedicine: the scarcity and privacy constraints of real patient data. Acquiring large and diverse sets of vascular geometries from clinical sources is costly, time-consuming, and often unfeasible due to confidentiality. Synthetic models offer a powerful solution by enabling the rapid generation of an unlimited number of diverse, anatomically plausible vascular structures. This capability is transformative for multiple applications, from basic research to clinical practice.

The relevance of these models is particularly evident in the field of simulations. Realistic virtual environments are essential for understanding complex biological processes and validating new medical technologies. For example, hemodynamic simulations—which analyze flow patterns, pressure, and shear stress—are fundamental for studying cardiovascular diseases such as atherosclerosis and aneurysms. Synthetic models make it possible to generate large datasets with variable geometries and pathologies, enabling large-scale studies that would be infeasible using only real data. This approach accelerates our understanding of how vessel shape influences disease progression and treatment response. By generating thousands of virtual aneurysms with different shapes and sizes, it becomes possible to systematically investigate which features correlate with a higher risk of rupture. Beyond basic research, synthetic models are crucial for the design and testing of medical devices. Devices such as stents, catheters, and heart valves must be rigorously evaluated across a wide range of anatomical configurations to ensure their safety and effectiveness.

Finally, these models have enormous potential for surgical planning and rehearsal. While patient-specific models are already used in complex surgeries, synthetic models can create a training environment for surgeons. By generating a broad spectrum of vascular anatomies and pathologies, our models provide a valuable resource for education, enabling experience with rare or complex cases in a virtual, risk-free setting. This improves skill acquisition and shortens the learning curve for new procedures.

6.2 Future directions

The first avenue for improvement targets the gap between standard generative metrics and the needs of vascular modeling. Set distances or image-overlap measures (e.g., MMD, COV, 1-NNA) are poor proxies for anatomical plausibility: two samples may be close in Chamfer distance and yet one of them may not be plausible. Future work could incorporate additional metrics to better assess the impact of the representation and provide a more comprehensive assessment of the quality of the generated structures.

As a complement, we propose task-based evaluation. Synthesized geometries should be assessed not only with geometric or topological scores, but also by their impact on clinically relevant problems: estimation of CFD-derived biomarkers, lesion detection, and planning of endovascular devices. This requires end-to-end benchmarks in which synthetic cohorts are meshed, simulated, and compared against real distributions of pressure, flow patterns, and shear-stress-based biomarkers, with uncertainty quantified at each stage. Evaluating the performance of synthesized vasculature in practical applications would provide a deeper understanding of its clinical and scientific relevance. Closing the synthetic–real gap will likely require adaptation and calibration strategies that align anatomical statistics (radii, tortuosity, branching angles) and physiological responses (e.g., pressure–flow relationships), together with training objectives aware of meshability and simulation that penalize non-watertight surfaces, poor element quality, or unrealistic boundary behavior.

A second direction is controllability. Clinically useful generation should be conditional on anatomy, physiology, and context: target territories, demographic covariates, comorbidities, device parameters, or hemodynamic goals. Conditioning mechanisms and latent controls can enable “design by intent,” where models produce vascular trees that satisfy specified perfusion demands, outlet resistances, or geometric constraints, and can be

steered toward rare phenotypes for data augmentation.

Topology remains a central challenge. Extending beyond tree structures to networks with cycles, anastomoses, and multiscale hierarchies demands representations that enforce connectivity, preserve luminal geometry, and remain compatible with meshing. Likewise, integrating domain-specific constraints or using physics-informed models could improve the plausibility and realism of generated structures in real-world scenarios. Natural next steps include hybrid pipelines that fuse graph priors (for branching logic and vessel ordering) with implicit or surface models (for sub-voxel detail), as well as topology-aware losses and constraints that stabilize training and prevent spurious breaks or mergers. Our approach shows promising advances in 3D vascular geometry synthesis, paving the way for improved clinical applications that could assist healthcare professionals in diagnosis, treatment planning, and surgical interventions.

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